

MIDTERM REVIEW

DORUK CETEMEN

Department of Economics and Financial Markets, Luiss Guido Carli University and EIEF

YONGGYUN KIM

Department of Economics, Florida State University

FEI LI

Department of Economics, University of North Carolina

CURTIS R. TAYLOR

Department of Economics, Duke University

We study why organizations conduct interim performance reviews when monetary rewards are limited. An interim review creates incentive capacity by allowing future work and career opportunities to serve as rewards for past performance. Optimal review policies map a continuum of performance outcomes into a simple incentive ladder: termination, tough or easy continuation, and, for exceptional performance, an early maximal reward with no further work. Review can even sustain high effort when terminal compensation alone cannot. Its timing balances two forces: waiting improves the information revealed by performance, but leaves less future work available to motivate the agent.

KEYWORDS: Feedback, bounded reward, incentive ladder, incentive capital.

Doruk Cetemen: dcetemen@luiss.it

Yonggyun Kim: ykim22@fsu.edu

Fei Li: lifei@email.unc.edu

Curtis R. Taylor: curtis.taylor@duke.edu

We thank Ralph Boleslavsky, Felix Feng, Canice Prendergast, Peter Norman Sørensen, Ming Yang, and John Zhu for helpful comments and suggestions. All remaining errors are our own.

1. INTRODUCTION

Performance reviews are ubiquitous in organizations and often have substantial consequences for an employee's career. Evaluations affect bonuses and merit pay, but also promotion, retention, and dismissal (Cappelli and Conyon, 2018). This raises a basic question: if an organization can ultimately observe an informative measure of performance and reward the employee accordingly, why evaluate performance before the end of the current phase of employment? In a standard moral-hazard model with unrestricted monetary transfers, there is no need to: it is optimal to bundle incentives from different stages into a sufficiently powerful terminal compensation scheme.

This paper shows that the answer changes fundamentally when monetary rewards are bounded. Let M denote the largest performance-contingent payment that the principal can make. The cap may be literal, but more broadly it captures organizations in which large financial bonuses are difficult, costly, or institutionally constrained. Such restrictions are particularly natural in public-sector employment, where pay is relatively compressed and promotion is an important source of incentives (Mas, 2017, Deserranno, Kastrau, and León-Ciliotta, 2025), and in nonprofit organizations, which make less use of output-contingent incentive pay than for-profit firms (DeVaro and Brookshire, 2007). More generally, many careers provide substantial incentives through advancement rather than immediate cash compensation. Tenure and promotion in academia and promotion to partnership in law, accounting, consulting, and other professional-service firms are prominent examples (Baker, Jensen, and Murphy, 1988, Barlevy and Neal, 2019, Levin and Tadelis, 2005). In these environments, the opportunity to continue on a favorable career path is itself a reward.

This observation gives interim review a role that is absent when monetary incentives are unbounded. A review allows the principal to allocate future opportunities according to past performance. An agent who performs well can be offered a more attractive continuation contract, while one who performs poorly can face a demanding continuation standard or termination. Rents that must be conceded to motivate future effort can do double duty: besides inducing future effort, they can be used as rewards for past effort. We refer to these future rents and continuation opportunities as *incentive capital*. When M is large, the principal can instead provide incentives directly with money and incentive capital has

little value. When M is tight, continuation opportunities become valuable because monetary rewards are scarce.

We study this mechanism in a canonical dynamic moral-hazard environment. A risk-neutral agent produces cumulative Gaussian output over a finite horizon and privately chooses costly effort.¹ A risk-neutral principal receives output net of compensation, which satisfies both limited liability and is bounded above by M . The principal may commit to one public interim review, choose when it occurs, and condition both compensation and the continuation opportunity on the review score. Past performance contains information about past effort but does not change the technology, productivity, or difficulty of future work. Thus review has no sorting, learning, or technological role in the baseline: any value it creates comes purely from incentives.

Our first main result is that a continuum of possible review scores is optimally mapped into at most four discrete actions. These actions form an *incentive ladder*, ordered by the rent they deliver to the agent. At the bottom is a *bad stop*: sufficiently poor performance leads to termination with no payment. The next rung is continuation under a difficult terminal standard, which induces future effort while leaving the agent relatively little rent. Better review performance earns continuation under an easier terminal standard and hence a larger rent. Finally, when first-phase incentives are sufficiently scarce, exceptional performance reaches a *good stop*: the agent receives the maximal payment M immediately and is released from the remaining workload. Thus termination can optimally occur at both tails of the performance distribution. The bad stop is the strongest punishment available; the good stop is the strongest reward available once money alone can no longer reward performance. Depending on parameters, two, three, or all four rungs are used.

The discreteness of this ladder is notable because both performance and the feasible set of continuation contracts are continuous. Its structure comes from the geometry of bounded incentive provision. Among contracts that induce future effort, the principal uses only the minimum- and maximum-rent continuation contracts. Together with the two extreme no-effort actions—termination without pay and termination with the maximal payment—these are the only exposed actions in the principal’s statewise assignment problem. Monotone

¹For ease of exposition, we assume that effort in the baseline model is chosen only twice, before the performance review and after it. We relax this assumption and allow the agent to choose effort continuously in Section 6.1.

likelihood ratios then order the four actions by review performance. In this sense, a continuous performance evaluation optimally produces a coarse rating system with only a small number of economically distinct consequences.

Our second set of results establishes when review is valuable. The payment bound M is again the key comparative static. When M is sufficiently large, no genuine review can improve on the terminal contract: uniformly over all review dates, the principal strictly prefers to wait until the end. As the cap tightens, however, the terminal contract exhausts its ability to provide incentives and continuation rents become increasingly valuable. Near the smallest cap at which full effort can be implemented without review, an interior review is always strictly profitable and all four rungs of the incentive ladder are used. Indeed, review can do more than reduce the cost of incentives. It can make high effort implementable when terminal evaluation alone cannot. At the no-review implementability boundary, splitting the project into two shorter incentive problems creates strict incentive slack in each phase. For payment caps slightly below that boundary, full effort is impossible under any terminal-only contract but can be induced with an interim review. Review therefore expands the organization's effective incentive capacity: future rents that would otherwise be an unavoidable cost of motivating later effort become an additional currency with which to motivate earlier effort.

Our third main result concerns *when* the review should occur. Delay has an obvious informational advantage. With more performance accumulated before the review, the distributions generated by working and shirking are easier to distinguish. But waiting also uses up the very future opportunities that make review valuable. A later review leaves less remaining work, a smaller range of continuation rents, and therefore less incentive capital with which to reward first-phase performance. Optimal review timing balances better information against more time left to motivate.

The endpoint results make this tradeoff particularly sharp. An arbitrarily early review is strictly harmful: in the Brownian model the right derivative of review value at date zero is minus infinity. A very short probationary period generates almost no useful information, while bounded rewards prevent the principal from cheaply magnifying that information into incentives. Every strictly optimal review is therefore bounded away from the initial date. At the other endpoint, when M is tight, moving the review slightly before the terminal date has first-order value because even a short continuation phase creates scarce incentive capital.

Thus the model predicts neither “review as soon as possible” nor “wait until performance is most informative.” The optimum is governed by the tension between learning about past effort and preserving future opportunities with which to reward it.

Finally, we examine three departures from the baseline. First, we allow the agent to choose effort at every instant; the four-rung ladder continues to apply, although the easy continuation standard becomes more demanding. Second, we study the opposite informational extreme in which output is publicly observed continuously and the agent can react to the entire path. The optimal contract then becomes a continuous-time band contract with firing and buyout boundaries. This benchmark also shows that more frequent review need not benefit the principal: over a substantial range of payment caps, committing to one optimally timed review dominates continuous observation. Finally, we show that random review timing can be valuable because it relaxes the sharp tradeoff between collecting more pre-review information and preserving post-review incentive capital.

Related Literature This paper contributes to the literature on interim performance evaluation and dynamic incentive provision by asking three questions: Why review? How should review information be used? And when to review?

First, we identify a new rationale for interim review. As in [Lizzeri, Meyer, and Persico \(2002\)](#), we consider a setting in which past performance does not change the primitive continuation agency problem. With unrestricted monetary rewards, the principal can bundle incentives across the two phases into a single terminal contract, making review unnecessary. However, when payments are bounded, as in the static setting of [Jewitt, Kadan, and Swinkels \(2008\)](#), review allows the principal to condition otherwise unavoidable continuation rents and the remaining workload on earlier performance, thereby creating history dependence and expanding incentive capacity. In contrast, much of the literature considers environments in which earlier actions or outcomes affect the difficulty or return of subsequent work. In such settings, an important role of interim review is to reveal the severity of the continuation agency problem and allow the continuation contract to respond accordingly (see, e.g., [Lukas, 2010](#), [Chen and Chiu, 2013](#), [Ely and Szydlowski, 2020](#)). A separate rationale is efficient sorting: [Ray \(2007\)](#) uses interim evaluation to identify and terminate projects with poor continuation prospects. Relatedly, [Liu \(2026\)](#) studies resource allocation when the agent privately knows a project’s fixed quality and always prefers a larger allocation. A

public midterm signal correlated with project quality allows the principal to discipline the agent's initial report, effectively "levering" information from the review into the first-period allocation. Our setting has neither private information about project quality to elicit nor heterogeneous continuation prospects to uncover. Instead, review provides information about the agent's hidden productive effort, and its value comes from using continuation decisions to provide incentives when monetary rewards are limited.

Second, we show that interim performance can be used through a simple, cutoff-based contract. Depending on interim performance, the agent is terminated or retained under a continuation contract that pays only following sufficiently good final performance. Termination can therefore occur at both tails: a bad stop punishes poor performance, while a good stop rewards strong performance by combining the maximal monetary payment with relief from future effort. Whereas [Ely, Georgiadis, and Rayo \(2025\)](#) jointly design dynamic incentives and the disclosure of performance feedback, our cutoff structure concerns how revealed interim performance is translated into termination and continuation incentives.

Good-performance termination also arises in other dynamic moral hazard settings (see, e.g., [Spear and Wang \(2005\)](#) and [Sannikov \(2008\)](#)), but for a different reason. In those models, favorable performance raises the promised value of a risk-averse agent until maintaining incentives becomes too costly, so termination occurs when continuation value reaches a sufficiently high level. Our good stop is instead a direct response to the payment cap. Once the maximal monetary prize has been reached, reducing the agent's remaining workload provides the additional reward that money cannot. This logic is closer to [Zhu \(2013\)](#), where periods of shirking can optimally serve as relaxation periods that reward good performance.

Third, we identify a new tradeoff governing the timing of review. Delaying review has a familiar benefit: more accumulated performance makes the review signal more informative. But delay has a novel cost in our setting: it shortens the remaining production horizon and thereby reduces the future output that can be used to create continuation rents. Review timing therefore determines not only the quality of information available to the principal, but also the amount of incentive capacity remaining after the review.

This question connects our paper to a recent literature on the design and timing of dynamic monitoring. [Varas, Marinovic, and Skrzypacz \(2020\)](#) show that optimal monitoring can take the form of deterministic periodic reviews when information acquisition is important, but random inspections when monitoring primarily serves to provide incentives. Similarly,

[Ball and Knoepfle \(2026\)](#) study whether inspections should be predictable or random in a long-term moral-hazard relationship, with the answer depending on whether effort primarily promotes breakthroughs or prevents breakdowns. [Wong \(2026\)](#) goes further by allowing the principal to design the monitoring technology itself, obtaining a nonstationary sequence of pass/fail tests whose informativeness and consequences evolve over the relationship. Relatedly, [Dai, Wang, and Yang \(2026\)](#) jointly design monitoring and compensation, with the principal dynamically reallocating limited monitoring capacity between evidence confirming effort and evidence revealing shirking. These papers endogenize the pattern, composition, or technology of monitoring, whereas we take a single interim review as given and ask where it should be placed within a finite production horizon.

Closest to our timing question, [Rodivilov \(2022\)](#) studies when to monitor an agent engaged in experimentation and shows that monitoring has both a static effect on contemporaneous moral hazard and a dynamic effect on the rents that must be provided in earlier periods. Our mechanism is different. In our model, the benefit of delay comes from the increasing informativeness of accumulated performance, while its cost comes from exhausting the very continuation opportunities through which review provides incentives.

This distinction also separates our mechanism from [Hoffmann, Inderst, and Opp \(2021\)](#) and [Georgiadis and Szentes \(2020\)](#), who study the timing of performance measurement following a one-time hidden action rather than an interim review during ongoing production. In [Hoffmann, Inderst, and Opp \(2021\)](#), delaying payout improves the information on which compensation is based but entails a direct cost of deferral; in [Georgiadis and Szentes \(2020\)](#), continued monitoring improves precision but incurs a direct monitoring cost. In our model, by contrast, the cost of delay arises from the contracting technology itself: waiting for a more informative review leaves less future work with which to reward favorable performance and provide incentives.

2. ENVIRONMENT

A risk-neutral principal (she) contracts with a risk-neutral agent (he) over a finite horizon $[0, T]$. Output X_t evolves as $dX_t = e_t dt + dW_t$, with $X_0 = 0$, where W is a standard

Brownian motion and $e_t \in \{0, 1\}$ is agent effort. We write $\Phi(\cdot)$ for the standard normal distribution and $\phi(\cdot)$ for its density.²

The contract fixes at date 0 a single review date $\tau \in (0, T]$.³ The review date is ultimately a choice variable, and we solve for it in two steps: Section 4 takes τ as *exogenous*, and Section 5 then *endogenizes* it.

ASSUMPTION 1—Observability of the output: *Cumulative output is publicly observed at τ and at T . No one, including the agent, observes X_t at any other time.*

The review splits the horizon into two phases, $P_1 = (0, \tau]$ and $P_2 = (\tau, T]$, of lengths τ and $T - \tau$; we call X_τ the *review score*. The case $\tau = T$ is the *no-review benchmark*: the second phase is empty and X_T is the only signal.

ASSUMPTION 2—Phase-level private effort: *Effort is constant within each phase. The agent privately chooses $a_1 \in \{0, 1\}$ at date 0 and, having observed X_τ , privately chooses $a_2 \in \{0, 1\}$ at date τ ; effort is $e_t = a_1$ on P_1 and $e_t = a_2$ on P_2 . A strategy is a pair (a_1, S) with $S : \mathbb{R} \rightarrow \{0, 1\}$ giving $a_2 = S(X_\tau)$.*⁴

The agent commits at the start of a phase to work throughout it or shirk throughout it. Consequently the two phase increments are independent given (a_1, a_2) , with

$$X_\tau \sim N(a_1\tau, \tau), \quad X_T - X_\tau \sim N(a_2(T - \tau), T - \tau).$$

Working through a phase of length h costs the agent ch , so total effort cost is $c[a_1\tau + a_2(T - \tau)]$ with $c \in (0, 1)$. The same flow cost applies for the entire horizon, and hence in both phases. After observing X_τ and X_T the principal makes a measurable payment $w(X_\tau, X_T)$ to the agent.

²Gaussianity is not essential for the main results. A general binary statistical experiment with suitable two-sided-evidence and monotone-likelihood-ratio conditions yield qualitatively similar findings. Details available upon request.

³The limiting case $\tau = 0$ is used only as a boundary limit in Section 5.

⁴Assumption 2 is relaxed in Section 6.1, where the agent chooses $e_t \in \{0, 1\}$ at each instant, so that a phase of length h delivers any total effort $\eta = \int e_t dt \in [0, h]$ rather than only $\{0, h\}$.

ASSUMPTION 3—Limited liability and finite compensation: *The payment function satisfies $w : \mathbb{R}^2 \rightarrow [0, M]$ with $M \in (0, \infty)$.*

The lower bound of 0 is the agent's limited liability, and the upper bound M is a finite cap: the maximum payment the principal can make. The principal receives ex post payoff $X_T - w$, and the agent receives $w - c[a_1\tau + a_2(T - \tau)]$. At $t = 0$ the principal chooses a contract that consists of a review date τ and a payment function w to maximize her expected payoff. Under the no-review benchmark $\tau = T$ a contract is simply $w(X_T) \in [0, M]$. The principal has all the bargaining power and full commitment, and the agent's outside option is zero.

3. CONTRACTING OVER A SINGLE PHASE

We begin the analysis with the single-phase contract. Fix a phase of length $h > 0$ at the end of which output is observed and payment is made, with no further review. Two cases of the model have this form, $h = T$ (no-review) and $h = T - \tau$ (post-review); in the latter, translation invariance of the technology lets a continuation contract be written as a function of the second phase output $Y = X_T - X_\tau$. Solving the phase problem once therefore delivers both cases.

Under work $Y \sim N(h, h)$ and under shirking $Y \sim N(0, h)$, with densities $f_j^h(y) = h^{-1/2}\phi((y - jh)/\sqrt{h})$ for $j \in \{0, 1\}$; working costs the agent ch . A contract is a measurable $w : \mathbb{R} \rightarrow [0, M]$; it implements work if

$$\mathbb{E}_1[w] - \mathbb{E}_0[w] \geq ch, \quad (\text{IC}_h)$$

where $\mathbb{E}_j[w] = \int w f_j^h$. Let $\mathcal{F}_h$ denote the set of contracts satisfying (IC_h) . Throughout the section c and h are fixed, and we suppress the dependence of derived objects on (h, c, M) except where an argument varies.

3.1. Threshold contracts

First, we show that it is without loss to focus on *threshold contracts*: pay the cap M when output clears a bar b and nothing below it. The reason is that the likelihood ratio $\ell^h(y) = f_0^h(y)/f_1^h(y) = \exp(h/2 - y)$ is strictly decreasing, so a contract that shifts payment

toward high output — rewarding evidence of work and withholding payment after evidence of shirking — provides incentives at the lowest expected cost.

LEMMA 1: *Let $w : \mathbb{R} \rightarrow [0, M]$ be any contract and let $\tilde{w} = \mathbf{1}_{\{y \geq b\}} M$ be the threshold contract with the same expected cost under work, $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$. Then $\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w]$, with equality only if $w = \tilde{w}$ almost everywhere. Hence $w \in \mathcal{F}_h$ implies $\tilde{w} \in \mathcal{F}_h$.*

Nothing is therefore lost by restricting attention to threshold contracts. Moreover, because every threshold contract pays either 0 or M , it can equivalently be interpreted as awarding a fixed prize M (e.g., a promotion) whenever performance clears the bar.

Now, index the bar b by the *normalized* value $z = (h - b)/\sqrt{h}$, so that a larger z is an easier bar, cleared with probability $\Phi(z)$ under work. For $w = \mathbf{1}_{\{y \geq b\}} M$,

$$\mathbb{E}_1[w] = M\Phi(z), \quad \mathbb{E}_0[w] = M\Phi(z - \sqrt{h}), \quad \mathbb{E}_1[w] - \mathbb{E}_0[w] = M\Delta(h, z), \quad (1)$$

where $\Delta(h, z) = \Phi(z) - \Phi(z - \sqrt{h})$ is the *incentive index*: the increase in the probability of clearing the bar when the agent works rather than shirks. By (1) a threshold contract implements work exactly when

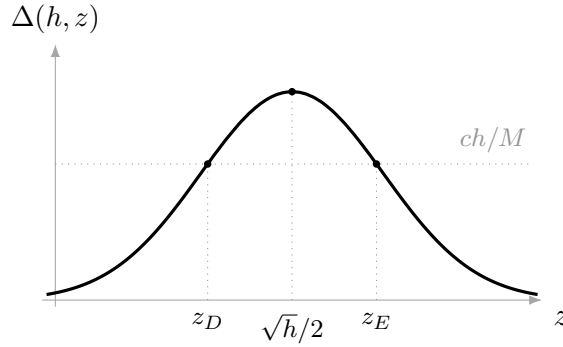
$$\Delta(h, z) \geq \frac{ch}{M}. \quad (2)$$

Figure 1 illustrates the admissible bars. The index is single-peaked at $z = \sqrt{h}/2$, where it attains its maximum $\bar{\Delta}(h) = 2\Phi(\sqrt{h}/2) - 1$, symmetric about that peak, and vanishes as $z \rightarrow \pm\infty$. Work is therefore implementable if and only if $M \geq M_I(h, c) \equiv \frac{ch}{\bar{\Delta}(h)}$ at every z . When it holds, the admissible bars form to the closed interval $[z_D, z_E]$ with

$$z_D = \min \left\{ z : \Delta(h, z) \geq \frac{ch}{M} \right\}, \quad z_E = \max \left\{ z : \Delta(h, z) \geq \frac{ch}{M} \right\}, \quad (3)$$

the two crossings of the horizontal line at height ch/M , which merge at $\sqrt{h}/2$ when $M = M_I(h, c)$. The constraint binds at z_D and z_E and is slack strictly between them. The lower crossing z_D corresponds to a high-output bar and sets a *difficult* standard b_D ; the upper crossing z_E corresponds to a low bar and sets an *easy* one b_E .

Implementability does not guarantee that the principal wants the agent to work: the rent she must concede to do so may exceed what the phase produces. A second threshold therefore governs profitability.

FIGURE 1.—The incentive index $\Delta(h, \cdot)$

LEMMA 2: *There is a threshold $\underline{M}(h, c) \geq M_I(h, c)$ such that work is implementable and the principal's payoff from implementing it is nonnegative if and only if $M \geq \underline{M}(h, c)$. Additionally, there exists $\bar{c}(h) \in (0, 1)$ such that $\underline{M}(h, c) = M_I(h, c)$ for all $c \leq \bar{c}(h)$; otherwise $\underline{M}(h, c) > M_I(h, c)$.*

Appendix A gives $\bar{c}(h)$ and $\underline{M}(h, c)$ in closed form. We impose the following assumption to ensure that contracting without a review is strictly profitable.

ASSUMPTION 4—Implementable and profitable full effort: $M > \underline{M}(T, c)$.

3.2. The no-review benchmark

Setting $h = T$ gives the benchmark against which a review must be evaluated: the principal writes a single contract on terminal output and never intervenes. By Lemma 1 she may take it to be a threshold contract, and among the admissible bars she chooses the cheapest, which is the difficult one.

PROPOSITION 1: *Under Assumption 4, the principal's optimal no-review payoff is $V^{\text{NR}}(T; c, M) = T - M\Phi(z_D(T, c, M)) > 0$, attained by the difficult bar. Moreover $z_D(T, c, M) \rightarrow -\infty$ and $V^{\text{NR}}(T; c, M) \rightarrow (1 - c)T$ as $M \rightarrow \infty$, so the first-best surplus accrues to the principal as the payment cap is relaxed.*

The limit is what makes a review worth studying. Since $z_D = (T - b_D)/\sqrt{T}$, z_D falling to $-\infty$ corresponds to a bar b_D rising to $+\infty$ in output: as M grows the principal promises an ever larger prize on an ever less likely outcome. Because the Gaussian likelihood ratio is

unbounded, that trade is favorable for the principal. The agent's rent is what he could secure by shirking, $\mathbb{E}_0[w] = M\Phi(z_D - \sqrt{T})$ by (1), and as the bar rises this vanishes even though M diverges: the reward event becomes overwhelming evidence of work. A principal who can promise arbitrarily large payments therefore captures the whole surplus without ever consulting interim performance, and has nothing to gain from a review. When M is finite the agent retains a strictly positive rent, and the question is whether the principal can do better by making the continuation depend on an interim signal.

3.3. The review action set

Now take $h = T - \tau$ to be a continuation phase. After a review score x the principal's choice is a continuation contract $w(x, \cdot) : \mathbb{R} \rightarrow [0, M]$ on the second-phase output Y , and the agent's best response to it determines whether second-phase work occurs. For everything that follows, such a contract is summarized by two numbers: the expected continuation utility r it leaves the agent net of effort cost, and the *total* expected continuation surplus s it generates—the principal's own expected continuation payoff is then $u = s - r$. We call the pair (r, s) an *action* and describe the review by the set of actions available at it, and drop the qualifier *expected* from payoffs and surplus below whenever there is no confusion. When effort is implemented, total surplus is $(1 - c)h$, so the only remaining question is which rents the agent can be promised.

LEMMA 3: *The rents attainable while implementing work are exactly $[r_D, r_E]$, where*

$$r_j = M\Phi(z_j) - ch = M\Phi(z_j - \sqrt{h}), \quad j \in \{D, E\}, \quad 0 < r_D < r_E < M - ch,$$

and $r_D + r_E = M - ch$.

Here r_D and r_E are the rents left by the difficult and the easy standard respectively. The principal's payoff from the difficult standard, $V^{\text{NR}} = (1 - c)h - r_D$, is the no-review benchmark value when $h = T$. The set of rent-surplus pairs available under work is therefore $\mathcal{A}_1 = [r_D, r_E] \times \{(1 - c)h\}$.

If work is not implemented, surplus is zero, and since feasible payments range over $[0, M]$ the corresponding set is $\mathcal{A}_0 = [0, M] \times \{0\}$. The feasible action set is therefore $\mathcal{A} = \mathcal{A}_0 \cup \mathcal{A}_1$. A review menu assigns one such action to every review score. Thus, after each score, the

principal chooses both a continuation profile—no work from $\mathcal{A}_0$ or work from $\mathcal{A}_1$ —and the rent delivered within that profile. The next section determines how these choices should vary with the review score in order to implement first-phase effort.

4. THE EXOGENOUS REVIEW DATE

Having established what can be offered after the review, we now move backward to the incentives that precede it. Throughout this section the review date $\tau \in (0, T)$ is fixed exogenously, and we continue to write $h = T - \tau$.

4.1. *The optimal incentive ladder*

The review performance is $X_\tau \sim N(a\tau, \tau)$ if the agent chose action $e_t = a \in \{0, 1\}$ throughout the first phase. Let g_1^τ and g_0^τ be the corresponding densities. Their likelihood ratio $g_1^\tau(x)/g_0^\tau(x) = \exp(x - \tau/2)$ is strictly increasing.

Let $a(x) = (r(x), s(x)) \in \mathcal{A}$ be the action assigned after review score x , delivering the agent rent $r(x)$ and the principal payoff $s(x) - r(x)$. Working through the first phase yields the agent $\int r g_1^\tau - c\tau$ and shirking yields $\int r g_0^\tau$, so pre-review obedience requires

$$\int_{\mathbb{R}} r(x) [g_1^\tau(x) - g_0^\tau(x)] dx \geq c\tau, \quad (\text{OB}_\tau)$$

which also guarantees participation, since $r \geq 0$ makes the utility from shirking nonnegative. Conditional on inducing first-phase work, expected output accumulated by the exogenous review is τ , so the principal's problem at $t = 0$ is

$$V^R(\tau) = \sup_{a(\cdot)} \tau + \int_{\mathbb{R}} [s(x) - r(x)] g_1^\tau(x) dx \quad (\text{P}_\tau)$$

subject to (OB_τ) , $a(\cdot)$ measurable, $a(x) \in \mathcal{A}$ for a.e. x .

PROPOSITION 2: Fix $\tau \in (0, T)$ and suppose first-phase work is implemented. An optimal menu takes the ordered form

$$a^*(x) = \begin{cases} A_B, & x < k_0^*, \\ A_D, & k_0^* \leq x < k_1^*, \\ A_E, & k_1^* \leq x < k_2^*, \\ A_G, & x \geq k_2^*, \end{cases}$$

where $A_B = (0, 0)$, $A_D = (r_D, (1 - c)h)$, $A_E = (r_E, (1 - c)h)$, $A_G = (M, 0)$ are four endpoints of $\mathcal{A}$, and $k_0^* \leq k_1^* \leq k_2^*$, strictly between any finite cutoffs.

Proposition 2 is our first main result. It shows that the optimal contract partitions the midterm-performance space into at most four ordered regions, assigning continuation profiles with progressively higher rents as performance improves. After the lowest performance realizations, A_B supplies the strongest punishment: the relationship ends with no payment and no second-phase production.⁵ At intermediate performance realizations the principal preserves the continuation surplus $(1 - c)h$. She first uses the difficult standard A_D , which delivers the smallest rent consistent with continuation effort, and then the easy standard A_E , which raises the agent's rent from r_D to r_E without sacrificing production. Only after the highest performance realizations can the principal move to A_G , paying the maximal rent M but giving up the continuation surplus altogether. This progression linking better review outcomes with higher discrete rent levels is what we refer to as the *incentive ladder*.

The optimal ladder therefore uses continuation contracts as intermediate incentive instruments. Before resorting to the maximal monetary reward, it increases the agent's continuation rent while keeping second-phase effort in place. The weak ordering of the cutoffs allows some of the upper profiles (rungs of the ladder) to be absent.

4.2. Deriving the optimal ladder

To understand why the optimal menu has this form, return to Problem ($\mathbf{P}_\tau$). This is an optimization over the entire menu $a(\cdot)$, and its only constraint linking choices across review

⁵Because both parties are risk-neutral, the outcome $A_B = (0, 0)$ can be implemented by either a hard stop in which the principal terminates the agent and x_t ceases to evolve, or a soft stop where the agent shirks out the clock.

scores is pre-review obedience. Despite its infinite-dimensional appearance, the problem admits an elementary geometric solution. The reduction also reveals the economic forces governing the choice among stopping and continuation arrangements.

Let $\lambda^* \geq 0$ be a Lagrange multiplier on (OB_τ) at the optimum and define the *net shadow value of continuation rent*

$$Q_\lambda(x) \equiv \lambda \left(1 - \frac{g_0^\tau(x)}{g_1^\tau(x)} \right) - 1 = \lambda \left[1 - \exp \left(\frac{\tau}{2} - x \right) \right] - 1. \quad (4)$$

Adjoining (OB_τ) and factoring out $g_1^\tau(x)$ gives the Lagrangian

$$\mathcal{L}(a; \lambda, \tau) = \tau - \lambda c\tau + \int_{\mathbb{R}} g_1^\tau(x) \{s(x) + Q_\lambda(x) r(x)\} dx. \quad (5)$$

For each λ , the Lagrangian is maximized over measurable menus with $a(x) \in \mathcal{A}$ for almost every x , and the optimal menu maximizes (5) at $\lambda = \lambda^*$, together with primal feasibility and complementary slackness. This reformulation separates the two agency problems: post-review incentive compatibility and payment feasibility are encoded in $\mathcal{A}$, while the multiplier brings pre-review obedience into the objective.⁶ Because the Lagrangian is additively separable across review scores, choosing a menu reduces to maximizing $s + Q_\lambda(x)r$ over $\mathcal{A}$ for almost every x , where $Q_\lambda(x)$ is the weight on the agent's continuation rent after score x . For $\lambda > 0$, $Q_\lambda(x)$ increases with x , as the work-to-shirking likelihood ratio does, so better performance tilts the choice toward higher-rent continuation profiles. By Lemma B.1(iii) in Appendix B, $\lambda^* > 0$ outside a knife-edge case, so this characterization applies at the optimum.

Although the original action choice at a given review score is from $\mathcal{A} = \mathcal{A}_0 \cup \mathcal{A}_1$, it is convenient to solve the linear program

$$a_\lambda(x) \in \arg \max_{(r,s) \in \text{co}(\mathcal{A})} \{s + Q_\lambda(x)r\}.$$

Because the objective is linear in (r, s) , convexification does not change the maximal value, and an optimizer can always be selected from the original set $\mathcal{A}$.

⁶This is the familiar likelihood-ratio formulation of a static moral-hazard problem (Holmström, 1979), except that the reward is a continuation profile rather than a wage.

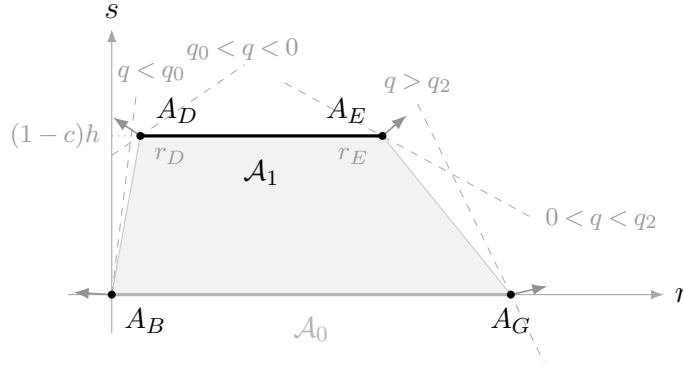


FIGURE 2.—The review action set and the supporting lines of the assignment objective

A linear objective over the compact polygon $\text{co}(\mathcal{A})$ is maximized at an extreme point, and its four extreme points A_B , A_D , A_E and A_G all belong to $\mathcal{A}$. Hence a maximizer can always be chosen from $\mathcal{A}$, and public randomization cannot improve the principal's payoff; at a breakpoint an entire edge may be optimal, but so are its endpoints. Figure 2 depicts this problem: the bold segments form $\mathcal{A}$, the shaded trapezoid is $\text{co}(\mathcal{A})$, and the dashed level sets of $s + qr$ are moved in the direction $(q, 1)$ until they support the hull.

Fix $q = Q_\lambda(x)$. The four vertices deliver

$$v_B(q) = 0, \quad v_D(q) = (1-c)h + qr_D, \quad v_E(q) = (1-c)h + qr_E, \quad v_G(q) = qM,$$

whose slopes $0 < r_D < r_E < M$ are ordered, so the upper envelope moves toward higher-rent actions as q rises. Adjacent vertices tie at

$$q_0 = -\frac{(1-c)h}{r_D} < q_1 = 0 < q_2 = \frac{(1-c)h}{M - r_E}, \quad (6)$$

where the level sets of $s + qr$, of slope $-q$, are parallel to the edges A_BA_D , A_DA_E and A_EA_G of Figure 2; concavity of the upper boundary of $\text{co}(\mathcal{A})$ gives the ordering. The maximizing action is therefore A_B for $q < q_0$, A_D for $q_0 < q < q_1$, A_E for $q_1 < q < q_2$, and A_G for $q > q_2$. Since $\lambda^* > 0$, $Q_{\lambda^*}(x)$ is strictly increasing in x , so as performance rises q passes through these intervals in order, yielding the ordered regions of Proposition 2.

4.3. On the form of the optimal ladder

We now ask which rungs the optimal ladder uses: when termination follows poor or strong performance, and when continuation is split into difficult and easy standards.

COROLLARY 1: Suppose $(1 - c)h \neq r_D$.⁷ Whenever first-phase work is implemented, k_0^* is finite and the bad-stop and difficult-standard regions are nonempty. Moreover:

- (i) If $1 + q_0 < \lambda^* \leq 1$, then $k_1^* = k_2^* = +\infty$, and only the bad-stop and difficult-standard regions are nonempty.
- (ii) If $1 < \lambda^* \leq 1 + q_2$, then $k_0^* < k_1^* < +\infty$ and $k_2^* = +\infty$, so the bad-stop, difficult-standard, and easy-standard regions are nonempty.
- (iii) If $\lambda^* > 1 + q_2$, then $k_0^* < k_1^* < k_2^* < +\infty$, and all four regions are nonempty.

The multiplier λ^* is the shadow value of the first-phase obedience constraint. It measures the principal's marginal gain from relaxing the incentive requirement for first-phase work. A large multiplier therefore means that this requirement is particularly costly to satisfy: first-phase incentives are scarce, and the principal is willing to use stronger rewards following favorable performance.

This interpretation is reflected directly in the menu. The index $Q_{\lambda^*}(x)$ sweeps the interval $(-\infty, \lambda^* - 1)$ as performance x ranges over $\mathbb{R}$, and it crosses breakpoint q_j if and only if $\lambda^* > 1 + q_j$. Implementing first-phase work requires $\lambda^* > 1 + q_0$, so the bad stop and the difficult standard are always used. Once $\lambda^* > 1$, the incentive value of additional rent is high enough to justify the easy standard, which rewards stronger performance while preserving second-phase production. Once $\lambda^* > 1 + q_2$, even the easy standard provides too little rent at the top: the principal then uses the good stop, paying the maximal reward despite sacrificing the continuation surplus.

The range of Q_{λ^*} also explains the asymmetry between the lower and upper regions. It is unbounded below, so sufficiently poor performance always receives the strongest punishment. Its upper bound is $\lambda^* - 1$, however, so the stronger rewards become active only when the shadow value of first-phase incentives is sufficiently large. Whenever a breakpoint is reached, the corresponding cutoff is the unique solution to $Q_{\lambda^*}(k_j^*) = q_j$.

An especially sharp implication emerges when the review occurs late. As τ approaches T , the remaining continuation horizon becomes short. The continuation rungs do not disappear, however. Instead, the threshold for activating the highest reward converges to its lower bound, while the shadow value of first-phase incentives remains strictly above that bound.

⁷The knife-edge case $(1 - c)h = r_D$, in which a zero supporting multiplier may arise, is discussed in Appendix B.

Consequently, every rung of the incentive ladder is used sufficiently close to the terminal date.

PROPOSITION 3: *Suppose Assumption 4 holds. Then $k_2^*(\tau) < +\infty$ for every review date τ sufficiently close to T from below.*

For any finite admissible cap, a sufficiently late review uses all four continuation profiles. To see the force behind this result, recall that the good stop becomes active when $\lambda^* > 1 + q_2$. As the continuation horizon $h = T - \tau$ vanishes, the surplus sacrificed by replacing the easy standard with the good stop shrinks faster than the additional rent thereby created. Hence $q_2(h) \rightarrow 0$. At the same time, the fixed-review shadow value converges to the no-review shadow value, which is strictly greater than one under Assumption 4. Thus the good-stop threshold is eventually crossed, and all four regions are active.

5. THE TIMING AND VALUE OF REVIEW

In this section, the principal chooses the review date endogenously from $\tau \in (0, T]$. A date $\tau \in (0, T)$ represents a genuine interim review. A review at $\tau = 0$ is equivalent to one at $\tau = T$, so we use $\tau = T$ to represent no review and do not consider $\tau = 0$ separately. We first study how the value of a genuine review varies with its date and then determine when such a review dominates no review.

5.1. The optimal genuine review date

For a genuine review date $\tau \in (0, T)$ with $h = T - \tau > 0$, define the continuation support function for a remaining horizon h and shadow value q by

$$\mathcal{V}(q, h) \equiv \max \{0, (1 - c)h + q r_D, (1 - c)h + q r_E, qM\}.$$

Complementary slackness implies that the optimized value equals the Lagrangian at the constrained optimum. Hence

$$V^R(\tau) = \tau - \lambda^* c \tau + \int_{\mathbb{R}} g_1^\tau(x) \mathcal{V}(Q_{\lambda^*}(x), T - \tau) dx,$$

where $Q_{\lambda^*}(x)$ is the net shadow value of continuation rent defined in Equation (4).

LEMMA 4: Suppose Assumption 4 holds. Then

$$\lim_{\tau \downarrow 0} V^R(\tau) = \lim_{\tau \uparrow T} V^R(\tau) = V^{\text{NR}}(T; c, M).$$

Both boundaries degenerate, for opposite reasons. As $\tau \downarrow 0$ the review score carries almost no information about first-phase effort, so there is nothing to condition on; as $\tau \uparrow T$ the continuation phase vanishes, so there is no continuation opportunity left to allocate. Either way the principal is left with the terminal contract. The no-review benchmark is therefore the right normalization across the whole interval, and any gain from review must be an interior phenomenon.

The tradeoff among genuine review dates. Away from knife-edge parameters, the constrained envelope theorem applies while holding the multiplier and the selected action fixed. The derivative decomposes as

$$\begin{aligned} \frac{dV^R(\tau)}{d\tau} = & \underbrace{(1 - \lambda^* c)}_{\text{first-phase production}} - \underbrace{\int_{\mathbb{R}} g_1^\tau(x) \frac{\partial \mathcal{V}}{\partial h}(Q_{\lambda^*}(x), h) dx}_{\text{incentive capital}} \\ & + \underbrace{\int_{\mathbb{R}} \left[\frac{\partial g_1^\tau(x)}{\partial \tau} \mathcal{V}(Q_{\lambda^*}(x), h) + g_1^\tau(x) \frac{\partial \mathcal{V}}{\partial q}(Q_{\lambda^*}(x), h) \frac{\partial Q_{\lambda^*}(x)}{\partial \tau} \right] dx}_{\text{information}}. \quad (7) \end{aligned}$$

The first term captures first-phase production. Delaying the review lengthens the first phase, adding a unit of production while tightening first-phase obedience, whose shadow value is charged at rate c . The second term is the opposing loss of incentive capital. A longer continuation phase is *incentive capital*: a family of work-inducing contracts that produce second-phase output while delivering different rents, so that rents conceded for second-phase incentives can be recycled into rewards for first-phase effort. As h falls this family shrinks, and strong first-phase rewards increasingly require sacrificing second-phase production, culminating in A_G . The third term is the information effect. It values a sharper performance measure: the review-score means under working and shirking differ by τ against a standard deviation $\sqrt{\tau}$, so their standardized separation grows with the review date; since the marginal value of the index is exactly the rent the menu assigns, precision is worth most at scores that already carry a large rent.

An interior optimum balances these three terms, setting (7) equal to zero. The optimal review date therefore does not merely trade an early noisy signal against a late precise one; it balances the precision of the review against the productive contracting capacity that survives it.

The suboptimality of extreme review dates. The next result distinguishes the implications of the review-date tradeoff near the two boundaries.

PROPOSITION 4: *Review dates sufficiently close to the initial date are strictly suboptimal. Review dates sufficiently close to the terminal date are also strictly suboptimal when the payment cap is sufficiently tight. Formally,*

- (i) *for every τ sufficiently close to the initial date, there exists another genuine review date τ' such that $V^R(\tau') > V^R(\tau)$;*
- (ii) *suppose $c \leq \bar{c}(T)$, so that $\underline{M}(T, c) = M_I(T, c)$. If M is sufficiently close to $\underline{M}(T, c)$ from above, then every review date τ sufficiently close to the terminal date is strictly dominated by an earlier genuine review date $\tau' < \tau$.*

The first conclusion reflects the information term in (7). A short first phase adds only a small amount of production and generates an extremely weak signal of effort. To motivate first-phase effort, the principal must nevertheless create enough dispersion in continuation rents across review outcomes. Because those rents are bounded, extracting the required incentives from an almost uninformative signal is disproportionately costly. In particular, the principal must fire the agent with positive probability based on vanishingly little information. The first-phase production gain therefore cannot compensate for the incentive loss generated by a review sufficiently close to the initial date.

The second conclusion reflects the opposing incentive-capital term. Near the terminal date, delaying the review yields a more informative score but leaves almost no continuation phase over which rents can be varied. A tight payment cap makes this loss especially costly: the terminal contract is already close to exhausting its incentive capacity, so continuation rents have a high shadow value. The marginal loss of incentive capital then dominates the production and information benefits of further delay, and an earlier review raises the principal's payoff.

As a result, under the tight-cap conditions of Proposition 4, if an optimal genuine review exists, it cannot lie near either the beginning or the end of the project. It is genuinely a *midterm* review. An implication is that short probationary periods observed in practice must therefore reflect motives absent from the model, such as screening an unknown type, learning match quality, or preserving an option to abandon a bad project.

5.2. When is a genuine review valuable?

We now compare genuine review with the option of no review. Define the review gain for $\tau \in (0, T)$ by $G(\tau) = V^R(\tau) - V^{\text{NR}}(T; c, M)$.

Loose payment caps. When the payment cap is loose, the no-review contract can concentrate compensation on an increasingly rare terminal outcome and implement effort with vanishing rent. A review cannot replicate this economy without either splitting the long experiment or relying on an almost uninformative early signal.

PROPOSITION 5: Fix $T > 0$ and $c \in (0, 1)$. There exists a finite $\bar{M}(T, c)$ such that $G(\tau) < 0$ for every $\tau \in (0, T)$ whenever $M > \bar{M}(T, c)$.

The result is uniform over review dates. A pointwise comparison would still allow the principal to move the review date as M rises and stay ahead of the benchmark. Proposition 5 rules this out. The rent compression established in Proposition 1 eventually dominates uniformly: no allocation of production between a review phase and a continuation phase can preserve both the precision and the production horizon of the terminal contract.

Tight payment caps. The comparison can reverse when the payment cap is sufficiently close to the implementability boundary. The next result records the resulting local value improvement.

PROPOSITION 6: Suppose $c \leq \bar{c}(T)$, so that $\underline{M}(T, c) = M_I(T, c)$. If M is sufficiently close to $\underline{M}(T, c)$ from above, there exists $\delta > 0$ such that

$$V^R(\tau) > V^{\text{NR}}(T; c, M) \quad \text{for every } \tau \in (T - \delta, T).$$

The logic combines the boundary value with the local comparison in Proposition 4. By Lemma 4, $G(\tau)$ converges to zero as τ approaches T from below. Under the stated tight-cap

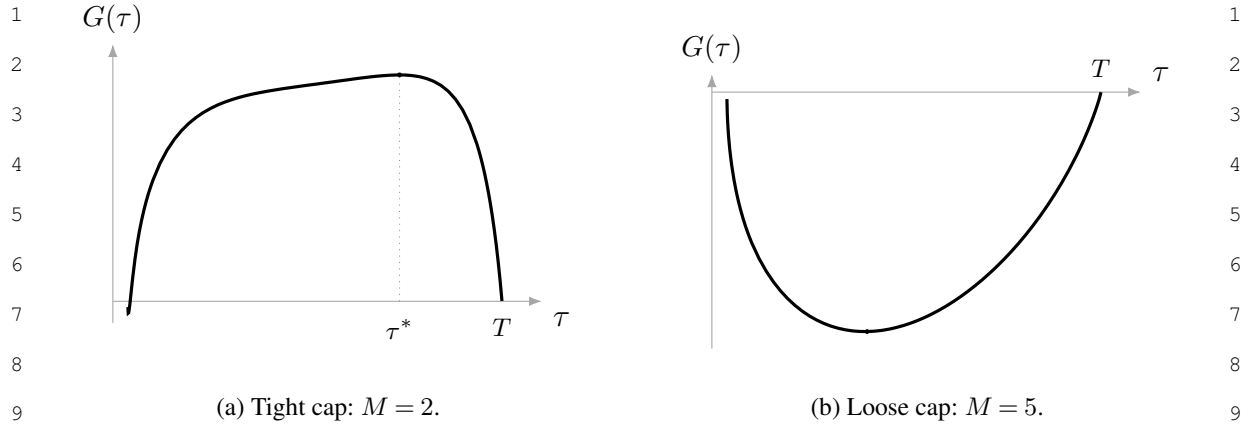


FIGURE 3.—The review gain $G(\tau)$ under a tight and a loose payment cap

conditions, Proposition 4 shows that moving a terminal-adjacent review earlier strictly raises $G(\tau)$. Together, these observations imply that $G(\tau) > 0$ for review dates sufficiently close to T , as stated in Proposition 6.

By Proposition 3, these profitable late reviews use all four rungs of the incentive ladder.

Payment caps and the value of review. Figure 3 illustrates the two regimes for $T = 1.5$ and $c = 0.6$. Under the tight cap $M = 2$, just above the implementability floor $M_I(T, c) = 1.96$, the gain reaches an interior maximum at $\tau^* = 1.09$, where $V^R = 0.239$ against $V^{NR} = 0.195$; the optimal menu uses all four rungs. Under the loose cap $M = 5$, the gain is negative at every review date and the principal waits until T .

Conflict over review. There is a simple conflict of interest over the use of review. Let $U^R(\tau)$ denote the agent's equilibrium payoff under the optimal contract with review at τ , and let U^{NR} denote his payoff under the optimal no-review contract. Since no review induces work throughout the horizon, it generates total surplus $(1 - c)T$. A review cannot increase total surplus and may reduce it by terminating production after some review outcomes. Hence

$$V^R(\tau) + U^R(\tau) \leq (1 - c)T = V^{NR} + U^{NR}.$$

It follows immediately that whenever review strictly benefits the principal, $V^R(\tau) > V^{NR}$, it strictly harms the agent: $U^R(\tau) < U^{NR}$. Thus any review that the principal would voluntarily introduce is opposed by the agent relative to no review.

Review restores implementability under tight caps. Under tight caps, the value of a midterm review also lies in restoring implementability. At the no-review implementability boundary, splitting the horizon creates strict incentive slack in both phases, and a review permits full effort for caps slightly below the level required without one.

PROPOSITION 7: *Let $M_0 \equiv M_I(T, c)$, at which full effort without review is just implementable. For every split $\tau \in (0, T)$, with $h = T - \tau$,*

$$M_0 \bar{\Delta}(\tau) > c\tau, \quad M_0 \bar{\Delta}(h) > ch.$$

Moreover there exist $\tau_0 \in (0, T)$ and $\varepsilon > 0$ such that for every $M \in (M_0 - \varepsilon, M_0)$ full effort is infeasible without a review, but a review at τ_0 admits a two-region menu inducing first-phase work and, whenever the project is continued, full continuation effort. Continuation occurs with positive probability.

A review can therefore create incentives unavailable when performance is evaluated only at the terminal date. Splitting the project gives two shorter incentive problems, each strictly easier to implement from $M_0 \bar{\Delta}(\tau) > c\tau$ and $M_0 \bar{\Delta}(h) > ch$. More importantly, it creates a new source of motivation: the continuation phase has a nondegenerate range of work-inducing rents $[r_D, r_E]$, and the principal can condition those rents on the review score. Rents that must in any event be conceded to motivate future effort thus double as rewards for past effort, supplemented when necessary by the two stopping actions.

6. EXTENSIONS

6.1. Pointwise effort choice

The baseline model restricts the agent to a phase-by-phase effort choice: in a phase of length h , he either works throughout or does not work at all. This section allows him to choose effort at each instant. Because no new information arrives within a phase, only total effort matters. Writing $e_t \in \{0, 1\}$ for instantaneous effort, let $\eta = \int_0^h e_t dt \in [0, h]$. Total effort η produces $Y \sim N(\eta, h)$ and costs $c\eta$. Thus the binary choice $\{0, h\}$ in the baseline becomes the interval $[0, h]$: the agent can now “coast” by working for only part of the phase.

This extension has two sharp implications. First, the four-rung structure of the optimal review menu survives: no interior effort level becomes an exposed continuation action.

Second, and more importantly, the easy continuation standard must become harder. Pointwise effort leaves the difficult continuation unchanged, but reduces the largest rent that the principal can deliver through the easy continuation. The extension therefore preserves the form of the incentive ladder while weakening one of its rungs.

From a global to a marginal incentive test. For a terminal wage w , let $\pi(\eta) = \mathbb{E}_\eta[w(Y)]$ denote expected compensation when total effort is η . A contract implements the target η exactly when no other effort level is more attractive:

$$\pi(\eta) - c\eta \geq \pi(\eta') - c\eta' \quad \text{for every } \eta' \in [0, h]. \quad (\text{IC}_h^{\text{flex}})$$

The phase-by-phase model checks only the extreme deviation $\eta' = 0$. Pointwise choice also introduces nearby ones, including the temptation to shave the last increment of effort. Two consequences of $(\text{IC}_h^{\text{flex}})$ carry the analysis: the comparison with $\eta' = 0$, and local optimality,

$$\pi(\eta) - c\eta \geq \pi(0), \quad (\text{G}_\eta)$$

$$\pi'(\eta) = c \text{ if } \eta \in (0, h), \quad \pi'(h) \geq c \text{ if } \eta = h. \quad (\text{L}_\eta)$$

The first is the baseline's *global* work-versus-shirk condition; the second is the new *marginal* one, an equality at an interior target and an inequality at full effort, where only downward deviations are available. For the Gaussian upper-tail bonus $w(y) = M\mathbf{1}_{\{y \geq h - z\sqrt{h}\}}$ these read $\Delta(h, z) \geq ch/M$ and $\sqrt{h}\phi(z) \geq ch/M$, and Online Appendix OA.1 shows that the two are jointly sufficient for this threshold contract to implement full effort.

The comparison with phase-by-phase effort is now immediate. At the difficult bar $z_D(h)$, the global condition binds and the marginal condition is slack. The difficult continuation is therefore unchanged. At the easy bar $z_E(h)$, by contrast, the baseline global condition binds but the marginal return to the last increment of effort is too small. The reason is simple: an easy bar is cleared with high probability even if the agent works slightly less, so coasting barely changes the probability of payment while saving effort cost. A contract can therefore induce “work rather than shirk” and still fail to induce “work all the way.”

The principal removes this temptation by raising the easy standard. In the z notation a larger z means an easier output bar, so the new bar is the positive root $\zeta(h)$ of

$$\sqrt{h}\phi(\zeta(h)) = \frac{ch}{M}, \quad \text{equivalently} \quad \phi(\zeta(h)) = \frac{c\sqrt{h}}{M}, \quad (8)$$

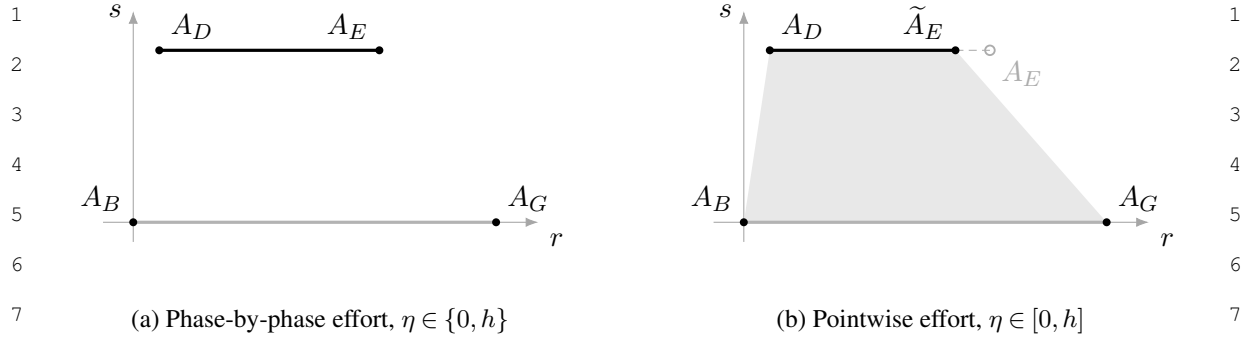


FIGURE 4.—Rent-surplus sets under phase-by-phase and pointwise effort.

which satisfies $\zeta(h) < z_E(h)$; the associated easy action is

$$\tilde{A}_E = (\tilde{r}_E(h), (1 - c)h), \quad \tilde{r}_E(h) = M\Phi(\zeta(h)) - ch < r_E(h). \quad (9)$$

Thus pointwise effort does not merely tighten full-effort feasibility; it specifically lowers the maximum continuation rent available at the easy end.

Figure 4 displays the central comparison. Moving from phase-by-phase to pointwise effort fills out the feasible rent-surplus set, but its convex hull still has four relevant vertices. Three are unchanged: bad stop A_B , difficult continuation A_D , and good stop A_G . Only the maximum-rent full-effort endpoint moves, from A_E to $\tilde{A}_E$. Geometrically, the full-effort face is shortened; economically, the easy continuation loses incentive capital.

The four-rung menu survives. Online Appendix OA.1 proves that the convex hull of $\mathcal{A}^{\text{flex}}(h)$ is $\text{co}\{A_B, A_D, \tilde{A}_E, A_G\}$: although pointwise effort permits a continuum of targets, no interior $\eta \in (0, h)$ maximizes the principal's statewise linear objective. The optimal menu can therefore still be chosen from four actions, ordered by review performance exactly as in the baseline—bad stop, difficult continuation, easy continuation, good stop—and all baseline formulas carry over after the economically meaningful substitution $r_E(h) \mapsto \tilde{r}_E(h)$. In particular the upper breakpoint becomes $\tilde{q}_2(h) = (1 - c)h / (M - \tilde{r}_E(h))$, while the lower breakpoint and the ordering of the four regions are unchanged. Since $\tilde{r}_E(h) < r_E(h)$, the easy continuation is a weaker reward: the principal uses it less and relies relatively more on the indivisible good-stop prize. Pointwise effort therefore changes the strength, but not the architecture, of the incentive ladder. This substitution away from continuation rents toward good stopping is costly: it reduces productive continuation and lowers the principal's payoff.

6.2. Continuous observability and the limit of frequent review

The baseline commits to a single public review date. A natural question is what happens when performance can instead be reviewed arbitrarily often. We study the limiting benchmark in which the entire output path is publicly observed. This changes more than the principal's information: because the agent also sees the path, effort can be conditioned on history. Thus continuous observability should be interpreted as the limit of increasingly frequent *public* reviews.

Output continues to satisfy $dX_t = e_t dt + dW_t$, but now $(X_s)_{s \leq t}$ is public at every date and $e_t \in \{0, 1\}$ is chosen pointwise. Let W_t denote the agent's promised continuation utility. Standard continuous-time incentive arguments (Sannikov, 2008) imply that, while work is recommended, promised utility must have sensitivity at least c to innovations in output. The principal therefore uses the smallest admissible sensitivity. The resulting contract has a simple band form.

PROPOSITION 8—Continuous-observation band contract: *Suppose output is publicly observed continuously and effort is chosen pointwise. An optimal contract can be represented by an initial promised utility $W_0^* \in [0, M]$ with the following properties.*

(i) *While $0 < W_t < M$, the agent works and promised utility evolves as*

$$dW_t = c dX_t, \quad \text{so that} \quad W_t = W_0^* + cX_t.$$

(ii) *The relationship ends when W_t first hits either boundary. At $W_t = 0$ the agent is terminated with zero payment; at $W_t = M$ he receives the maximal payment M and is released from further work. If neither boundary is reached before T , he is paid W_T at the horizon.*

(iii) *Writing $u = W_0/c$, $m = M/c$, and letting $v(T, u)$ be the expected time before T that a unit-drift Brownian motion starting at u remains inside $(0, m)$, the principal chooses*

$$u^* \in \arg \max_{u \in [0, m]} \{(1 - c)v(T, u) - cu\}.$$

The optimum is unique whenever it is interior.

The contract is the continuous-information analogue of the incentive ladder. The lower and upper absorbing boundaries play the roles of the bad and good stops. Between them,

however, there is no longer a pair of discrete continuation rents: every instant acts as a review, so the agent's continuation utility moves continuously with performance. The slope c is the minimum sensitivity consistent with effort, and the only initial design choice is how much "cushion" W_0^* to place between the two stopping boundaries. Appendix D proves Proposition 8; Online Appendix OA.2 gives the fuller recursive derivation and numerical details.

This limiting benchmark delivers an enlightening answer to the multiple-review question. More observation is not necessarily better for the principal, because the agent's ability to react to the running score tightens incentive compatibility history by history. At the benchmark used in Figure 3, $T = 1.5$, $c = 0.6$, and $M = 2$, the optimal single review occurs at $\tau^* = 1.089$ and gives the principal 0.239, while the continuous-observation band contract gives only 0.135; no review gives 0.195. The band contract actually leaves the agent less rent than either discrete-review contract at these parameters, but it induces only about 0.854 units of expected work because the lower boundary is hit frequently. The loss is therefore not due to excessive rent but inefficient early termination.

The comparison varies with the payment cap. Continuous observation is attractive when M is very tight because effort can be purchased in small increments and the relationship can stop as soon as promised utility reaches a boundary. Once the cap is moderate, however, a single review strictly dominates the band contract; with a sufficiently loose cap, the optimal single-review contract itself converges to no review. For $T = 1.5$ and $c = 0.6$, the band is profitable for caps above approximately 0.670 and dominates the best contract with at most one review only up to approximately $M = 1.488$. The no-review contract becomes implementable at approximately $M = 1.958$. Thus continuous observability is most valuable precisely in the region where coarse discrete evaluation has too little incentive capacity.

6.3. Random review timing with pointwise effort

The review date has so far been announced at date 0. We now ask whether uncertainty about timing can itself be useful when effort is chosen pointwise. For tractability, the principal commits to two possible dates $0 < \tau_1 < \tau_2 < T$, with probability p on the early date, but does not reveal the realization until a review occurs. Before τ_1 the agent therefore does not know which date will apply. If no review occurs at τ_1 , he learns that the review will

occur at τ_2 and can reoptimize his remaining effort. Thus the lottery must remain incentive compatible after the information conveyed by non-arrival.

Put $d = \tau_2 - \tau_1$, and let r_i be the rent schedule used if review occurs at τ_i . If cumulative effort before the first possible review is e , define

$$U_i(e) = \mathbb{E}[r_i(e + \sqrt{\tau_i}Z)] - ce, \quad \hat{U}_2(e) = \max_{m \in [e, e+d]} U_2(m), \quad Z \sim N(0, 1), \quad (10)$$

where $\hat{U}_2(e)$ is the late-branch value after the agent has inferred the review date and optimally revised his effort. Let $D_1(e) = U_1(\tau_1) - U_1(e)$ and $D_2(e) = U_2(\tau_2) - \hat{U}_2(e)$.

PROPOSITION 9: *Suppose Assumption 4 holds and effort is chosen pointwise. Fix $0 < \tau_1 < \tau_2 < T$ and put probability p on τ_1 .*

(i) *Full pre-review effort is sequentially incentive compatible if and only if $U_2(\tau_2) = \hat{U}_2(\tau_1)$ and*

$$pD_1(e) + (1 - p)D_2(e) \geq 0 \quad \text{for every } e \in [0, \tau_1]. \quad (11)$$

(ii) *If the late branch induces full effort on its own, then (11) holds if and only if $p \leq \bar{p}$, where*

$$\bar{p} = \inf_{\{e: D_1(e) < 0\}} \frac{D_2(e)}{D_2(e) - D_1(e)}, \quad (12)$$

with $\bar{p} = 1$ when the set is empty.

(iii) *Let P_i be the principal's payoff on branch i under full effort and let $V^{D, \text{pw}}$ denote the best deterministic value among contracts inducing full pointwise effort. If $P_1 > V^{D, \text{pw}} > P_2$, a two-date lottery strictly dominates every such deterministic contract whenever*

$$\underline{p} \equiv \frac{V^{D, \text{pw}} - P_2}{P_1 - P_2} < \bar{p}; \quad (13)$$

any $p \in (\underline{p}, \bar{p}]$ then delivers the improvement.

The attractive early branch can tolerate only so much weight before some early coasting becomes profitable; this gives the ceiling $\bar{p}$. Beating the best deterministic review requires enough weight on that branch; this gives the floor $\underline{p}$. Random timing is valuable when the incentive ceiling lies above the payoff floor.

A numerical example shows that the interval can be nonempty. Set $T = 1.5$, $c = 0.72$ and $M = 2.60$, and take $\tau_1 = 0.094868$ and $\tau_2 = 0.816614$. Using the pointwise-feasible continuation actions $A_B, A_D, \tilde{A}_E, A_G$ from Section 6.1, the branch payoffs are $P_1 = 0.13301$ and $P_2 = 0.03199$. The late branch induces full effort on its own and $\bar{p} = 0.47293$. A relaxation of the deterministic pointwise problem gives the upper bound $V^{D,pw} \leq 0.07850435$, and hence $\underline{p} \leq 0.46047 < \bar{p}$. At $p = 0.465$ the lottery is sequentially incentive compatible and yields 0.07896, strictly above that upper bound and therefore above every deterministic full-effort pointwise review.

The gain is small and the example is a witness rather than an optimized lottery. Its point is qualitative: even when the agent can coast before the first possible date, learns from the non-arrival of a review, and then optimally revises his effort, uncertainty about review timing can strictly improve on every deterministic review. Appendix D proves the proposition, and Online Appendix OA.3 gives the numerical construction.

7. CONCLUSION

This paper studies how a principal uses a public midterm review when effort is sequential and compensation is bounded. The review does more than evaluate past performance: it allows the principal to use future production, continuation rents, and termination as incentive instruments. The optimal menu assigns these continuation opportunities through an ordered performance ladder, while the review date balances the informativeness of accumulated performance against the incentive value of the remaining continuation phase.

The extensions also clarify the scope of the one-review institution. Pointwise effort preserves the ordered menu but weakens the easy-continuation rung. At the opposite extreme, continuous public observation produces a band contract and shows that arbitrarily frequent review need not dominate a single optimally timed review: finer feedback also gives the agent finer opportunities to condition his effort. Random review timing provides yet another margin by pooling incentive slack across possible dates. Thus, when incentives are dynamic, the principal may benefit not from observing performance as often as possible, but from choosing carefully when to look.

APPENDIX A: PROOFS FOR SECTION 3

Recall $\bar{\Delta}(h) = \max_z \Delta(h, z) = 2\Phi(\sqrt{h}/2) - 1$ and $M_I(h, c) = ch/\bar{\Delta}(h)$ from Section 3.1, and put $\bar{c}(h) = [2\Phi(\sqrt{h}/2) - 1]/\Phi(\sqrt{h}/2)$. Write

$$\ell_T = \frac{\phi(z_D(T) - \sqrt{T})}{\phi(z_D(T))} = \exp\left(z_D(T)\sqrt{T} - \frac{T}{2}\right), \quad \lambda_T^{\text{NR}} = \frac{1}{1 - \ell_T}, \quad (\text{A.1})$$

for the likelihood ratio at the no-review difficult bar and the shadow price of the no-review incentive constraint. Assumption 4 gives $z_D(T) < \sqrt{T}/2$, hence $\ell_T \in (0, 1)$ and $\lambda_T^{\text{NR}} > 1$.

PROOF OF LEMMA 1: Let $k \in [-\infty, +\infty]$ solve $\mathbb{P}\{Y \geq k \mid e = 1\} = \mathbb{E}_1[w]/M$, which exists and is unique because that probability falls continuously and strictly from one to zero (with the endpoints understood as limits), and write $\varphi = w/M$, $\varphi^* = \mathbf{1}_{\{y \geq k\}}$. Since $\int (\varphi - \varphi^*) f_1^h = 0$, subtracting $\ell^h(k)$ times this vanishing integral gives

$$\mathbb{E}_0[w] - \mathbb{E}_0[\tilde{w}] = M \int (\varphi - \varphi^*) [\ell^h - \ell^h(k)] f_1^h.$$

For $y > k$, $\varphi^* = 1 \geq \varphi$ and $\ell^h(y) < \ell^h(k)$; for $y < k$ both inequalities reverse. The integrand is therefore nonnegative almost everywhere, so $\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w]$, with equality only if $\varphi = \varphi^*$ a.e. Subtracting from $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$ gives the incentive comparison. *Q.E.D.*

PROOF OF LEMMA 2: *Implementability.* By Lemma 1 the largest attainable incentive spread is $M\bar{\Delta}(h)$, so $\mathcal{F}_h \neq \emptyset$ exactly when $M \geq M_I(h, c)$, and $[z_D, z_E]$ is nondegenerate exactly when the inequality is strict. With $a = \sqrt{h}/2$ we have $\bar{\Delta}(h)/h = \mathcal{G}(a) \equiv [2\Phi(a) - 1]/(4a^2)$ and $\mathcal{G}'(a) = -F(a)/(2a^3)$ with $F(a) = 2\Phi(a) - 1 - a\phi(a)$; since $F(0) = 0$ and $F'(a) = (1 + a^2)\phi(a) > 0$, $\mathcal{G}$ is strictly decreasing, so $M_I = c/\mathcal{G}$ is strictly increasing in h . *Profitability.* By Lemma 1 and (1) the cheapest work-inducing contract is the difficult bar, so the principal's payoff is $V(M) = h - M\Phi(z_D)$. Dividing the binding condition $M[\Phi(z_D) - \Phi(z_D - \sqrt{h})] = ch$ by $\Phi(z_D)$,

$$M\Phi(z_D) = \frac{ch}{1 - q(z_D)}, \quad q(z) = \frac{\Phi(z - \sqrt{h})}{\Phi(z)}. \quad (\text{A.2})$$

Since ϕ/Φ is strictly decreasing, $\frac{d}{dz} \log q > 0$, so q is a strictly increasing bijection onto $(0, 1)$. As M rises, ch/M falls and z_D lies on the increasing branch of $\Delta(h, \cdot)$, so z_D falls, hence

by (A.2) so does $M\Phi(z_D)$: the payoff V is strictly increasing in M , with $V \uparrow (1-c)h > 0$ as $M \rightarrow \infty$ and, as $M \downarrow M_I(h, c)$ where the two bars merge at $\sqrt{h}/2$, $V \rightarrow h(1 - c/\bar{c}(h))$. *The threshold.* If $c \leq \bar{c}(h)$ then $h(1 - c/\bar{c}(h))$ is nonnegative and $V > 0$ at every $M > M_I(h, c)$; set $\underline{M}(h, c) = M_I(h, c)$. If $c > \bar{c}(h)$ it is negative, so by strict monotonicity there is a unique $M_P(h, c) > M_I(h, c)$ with $V = 0$, and $\underline{M}(h, c) = M_P(h, c)$. At that cap $M\Phi(z_D) = h$ while the binding constraint gives $M\Phi(z_D - \sqrt{h}) = (1-c)h$; dividing, z_D solves $q(z) = 1 - c$, and M_P is h over Φ of that root. In both cases $\underline{M} \geq M_I$, with equality exactly when $c \leq \bar{c}(h)$. *Q.E.D.*

PROOF OF PROPOSITION 1: By Lemma 1 and (1) the cheapest work-inducing contract is the difficult bar, whose expected cost under work is $M\Phi(z_D(T))$; thus, $V^{\text{NR}}(T; c, M) = T - M\Phi(z_D(T, c, M))$. Positivity follows from Lemma 2 under Assumption 4. The limits of V^{NR} and $z_D(T) \rightarrow -\infty$ are established in the proof of Lemma 2. *Q.E.D.*

PROOF OF LEMMA 3: Take $w \in \mathcal{F}_h$ and let $\tilde{w}$ be the cost-matched threshold contract of Lemma 1, which also lies in $\mathcal{F}_h$; its bar therefore satisfies (2), so $z \in [z_D, z_E]$ and $M\Phi(z_D) \leq \mathbb{E}_1[w] \leq M\Phi(z_E)$. Both bounds are attained, and $r = \mathbb{E}_1[w] - ch$ is affine in w on the convex set $\mathcal{F}_h$, so the attainable rents are exactly $[r_D, r_E]$, swept by mixtures of the two extreme contracts. The second equality is (1) at a binding constraint, and $r_E < M - ch$ because $\Phi(z_E) < 1$. Finally $\Delta(h, z) = \Phi(z) + \Phi(\sqrt{h} - z) - 1$ is symmetric about $\sqrt{h}/2$, so $z_E = \sqrt{h} - z_D$ and $r_E = M\Phi(-z_D) = M - M\Phi(z_D) = M - ch - r_D$. *Q.E.D.*

The next lemma records two properties of the extreme continuation rents that are used repeatedly once the horizon is allowed to vary: they move continuously with the length of the continuation phase, and they spread to the full interval $[0, M]$ as that phase vanishes.

LEMMA A.1: Suppose Assumption 4 holds. Then $M\bar{\Delta}(h) > ch$ for every $h \in (0, T]$; the bars $z_D(h)$ and $z_E(h)$ of (3) are continuously differentiable in h there; and r_D and r_E are continuous on $(0, T]$, with $r_D(h) \rightarrow 0$ and $r_E(h) \rightarrow M$ as $h \downarrow 0$. Consequently $h \mapsto \mathcal{A}(h)$ is continuous on $(0, T]$ in the Hausdorff metric, and $\mathcal{A}(h) \rightarrow \mathcal{A}(0) = \{(r, 0) : r \in [0, M]\}$ as $h \downarrow 0$.

PROOF: *Strict feasibility.* Assumption 4 and Lemma 2 give $M > \underline{M}(T, c) \geq M_I(T, c)$, hence $M\bar{\Delta}(T) > cT$. Since $\bar{\Delta}(h)/h$ is strictly decreasing, as shown in the proof of Lemma 2,

every $h \in (0, T)$ satisfies $\bar{\Delta}(h)/h > \bar{\Delta}(T)/T > c/M$. In particular the two roots are separated: $z_D(h) < \sqrt{h}/2 < z_E(h)$.

Continuity. Fix $h \in (0, T]$. Because $\Delta(h, \cdot)$ is strictly increasing to the left of $\sqrt{h}/2$ and strictly decreasing to its right, the partial derivative $\partial_z \Delta(h, z) = \phi(z) - \phi(z - \sqrt{h})$ is strictly positive at $z_D(h)$ and strictly negative at $z_E(h)$. Since Δ is C^1 in (h, z) , the implicit function theorem applied to $\Delta(h, z) = ch/M$ makes z_D and z_E continuously differentiable at h . By Lemma 3, $r_j(h) = M\Phi(z_j(h) - \sqrt{h})$ is then continuous.

Short-horizon limits. Let $h \downarrow 0$. Applying the mean value theorem to $\Delta(h, z) = \Phi(z) - \Phi(z - \sqrt{h})$ at the binding difficult bar yields $\xi_h \in (z_D(h) - \sqrt{h}, z_D(h))$ with $M\sqrt{h}\phi(\xi_h) = M\Delta(h, z_D(h)) = ch$, that is, $\phi(\xi_h) = c\sqrt{h}/M \rightarrow 0$, so $|\xi_h| \rightarrow \infty$. As $\xi_h < z_D(h) < \sqrt{h}/2 \rightarrow 0$, the family is eventually bounded above, so $\xi_h \rightarrow -\infty$; and $z_D(h) < \xi_h + \sqrt{h}$ then gives $z_D(h) \rightarrow -\infty$. Hence $r_D(h) = M\Phi(z_D(h) - \sqrt{h}) \leq M\Phi(z_D(h)) \rightarrow 0$, and by Lemma 3, $r_E(h) = M - ch - r_D(h) \rightarrow M$.

The action set. $\mathcal{A}_0 = [0, M] \times \{0\}$ does not depend on h , while $\mathcal{A}_1(h) = [r_D(h), r_E(h)] \times \{(1 - c)h\}$ is a segment whose endpoints and height all vary continuously in h ; Hausdorff continuity of the union follows. As $h \downarrow 0$ that segment converges to $[0, M] \times \{0\}$, which is $\mathcal{A}_0$. *Q.E.D.*

APPENDIX B: PROOFS FOR SECTION 4

Review-score cutoffs and incentive supply

Section 4.2 establishes that, for a given $q = Q_\lambda(x)$, the maximizing action moves through A_B, A_D, A_E, A_G at the breakpoints $q_0 < q_1 = 0 < q_2$ in (6). For every $\lambda > 0$, Q_λ is strictly increasing in x and maps $\mathbb{R}$ onto $(-\infty, \lambda - 1)$. Hence the cutoff associated with q_j is finite exactly when $\lambda > 1 + q_j$, and inverting $Q_\lambda(k) = q_j$ gives

$$k_j(\lambda) = \frac{\tau}{2} - \log \left(1 - \frac{1 + q_j}{\lambda} \right) \quad \text{if } \lambda > 1 + q_j, \quad k_j(\lambda) = +\infty \quad \text{otherwise.} \quad (\text{B.1})$$

For later use, define $\underline{\lambda}(h) \equiv \max\{0, 1 + q_0\} = \max\{0, -V^{\text{NR}}(h)/r_D\}$, where $V^{\text{NR}}(h) = (1 - c)h - r_D$. Moreover, $1 + q_2 = (h + r_D)/(ch + r_D) < 1/c$, so any supporting multiplier $\lambda^* \geq 1/c$ makes all four regions active.

Define the standardized cutoff under first-phase work by $z_j(\lambda, \tau) \equiv (\tau - k_j(\lambda))/\sqrt{\tau}$. By (B.1),

$$z_j(\lambda, \tau) = \frac{\sqrt{\tau}}{2} + \frac{1}{\sqrt{\tau}} \log \left(1 - \frac{1 + q_j}{\lambda} \right), \quad j \in \{0, 1, 2\}, \quad (\text{B.2})$$

with $z_j = -\infty$ when $k_j = +\infty$. The induced rent schedule has jumps r_D , $r_E - r_D$, and $M - r_E$ at k_0, k_1, k_2 , so its first-phase incentive supply is

$$I(\lambda; \tau) = r_D \Delta(\tau, z_0) + [r_E - r_D] \Delta(\tau, z_1) + [M - r_E] \Delta(\tau, z_2), \quad (\text{B.3})$$

where $\Delta(\tau, -\infty) = 0$. Since each tie set $\{x : Q_\lambda(x) = q_j\}$ is null, (B.3) is unaffected by how ties are resolved.

The supporting multiplier

LEMMA B.1: Fix $\tau \in (0, T)$ and suppose Assumption 4 holds.

- (i) Monotonicity and range. $I(\lambda; \tau)$ is continuous and weakly increasing on $\lambda > 0$, and strictly increasing whenever the selected rent changes on a set of positive probability. Moreover, $\lim_{\lambda \rightarrow \infty} I(\lambda; \tau) = M\bar{\Delta}(\tau)$. If $(1 - c)h \neq r_D$, then $\lim_{\lambda \downarrow \underline{\lambda}(h)} I(\lambda; \tau) = 0$.
- (ii) Existence. $M\bar{\Delta}(\tau) > c\tau$, and Problem (P_τ) admits a supporting multiplier $\lambda^* \geq 0$ satisfying complementary slackness.
- (iii) Positivity and binding obedience. If $(1 - c)h \neq r_D$, every supporting multiplier satisfies $\lambda^* > 0$, and pre-review obedience binds:

$$\int_{\mathbb{R}} r^*(x) [g_1^\tau(x) - g_0^\tau(x)] dx = c\tau, \quad (\text{B.4})$$

where r^* is the rent delivered by the optimal menu. At the knife edge $(1 - c)h = r_D$, a zero multiplier may also support an optimum.

- (iv) Characterization. If $(1 - c)h \neq r_D$, every supporting multiplier solves $I(\lambda; \tau) = c\tau$. The solution set is a nonempty compact interval contained in $(\underline{\lambda}(h), \infty)$, and is a singleton whenever $I(\cdot; \tau)$ is strictly increasing at that level.

PROOF: (i) Let $\lambda'' > \lambda'$ and let (r', s') and (r'', s'') be Lagrangian maximizers at a given review score. Write $d_\tau(x) = 1 - g_0^\tau(x)/g_1^\tau(x)$. Adding the two optimality inequalities for $s - r + \lambda d_\tau r$ gives $(\lambda'' - \lambda')d_\tau(x)[r''(x) - r'(x)] \geq 0$. Multiplying by g_1^τ and integrating yields monotonicity of I , with strictness under the stated condition. Continuity follows

from (B.2): when a cutoff first becomes finite, its standardized value tends to $-\infty$ and its contribution to (B.3) vanishes.

If $(1 - c)h \neq r_D$, the limiting assignment becomes constant across review scores. When $1 + q_0 > 0$, it converges to A_B ; when $1 + q_0 < 0$, it converges to A_D . In either case $I \rightarrow 0$. Finally, as $\lambda \rightarrow \infty$, every $k_j \rightarrow \tau/2$, so every $z_j \rightarrow \sqrt{\tau}/2$; since the three rent jumps sum to M , $I \rightarrow M\bar{\Delta}(\tau)$.

(ii) By Lemma 2 and Assumption 4, $M > \underline{M}(T, c) \geq M_I(T, c) > M_I(\tau, c) = c\tau/\bar{\Delta}(\tau)$, where the strict inequality uses $\tau < T$ and monotonicity of $M_I(\cdot, c)$. Thus pre-review obedience is strictly feasible. Slater's condition and Kuhn–Tucker yield a supporting $\lambda^* \geq 0$ with complementary slackness.

(iii) Suppose $\lambda^* = 0$. Then $Q_0 \equiv -1$, so the review-score objective is $s - r$ and does not depend on x . If $(1 - c)h \neq r_D$, its maximizer is unique: A_B when $(1 - c)h < r_D$ and A_D when $(1 - c)h > r_D$. The assigned rent is therefore constant across scores and supplies no first-phase incentives, contradicting $c\tau > 0$. Hence $\lambda^* > 0$, and complementary slackness gives (B.4).

At $(1 - c)h = r_D$, instead, $q_0 = -1$, so under $\lambda = 0$ the bad stop A_B and difficult standard A_D are tied at every review score, while A_E and A_G are strictly dominated. A score-dependent selection between the tied actions can therefore provide the required first-phase incentives, so a zero supporting multiplier cannot be ruled out.

(iv) By (iii), $\lambda^* > 0$, so binding obedience is exactly $I(\lambda; \tau) = c\tau$. Parts (i)–(ii) imply that its solution set is nonempty, compact, and bounded away from $\underline{\lambda}(h)$. Weak monotonicity makes the level set an interval, and strict monotonicity at the level makes it a singleton.

Q.E.D.

The full ladder near the terminal date

The terminal boundary also yields a useful characterization of the shape of the fixed-review menu. The argument uses two limits. First, the threshold for activating the good stop converges to one. Second, the fixed-review supporting multiplier converges to the shadow price of the terminal no-review problem, which is strictly larger than one.

LEMMA B.2: Suppose Assumption 4 holds and write $h = T - \tau$. Then, as $h \downarrow 0$,

$$\frac{r_D(h)}{h} \longrightarrow +\infty, \quad q_2(h) \longrightarrow 0, \quad \lambda^*(T - h) \longrightarrow \lambda_T^{\text{NR}} > 1,$$

with ℓ_T and λ_T^{NR} as in (A.1).

PROOF: For the first limit, the difficult continuation bar satisfies $M[\Phi(z_D(h)) - \Phi(z_D(h) - \sqrt{h})] = ch$. By the mean-value theorem there is $\xi_h \in (z_D(h) - \sqrt{h}, z_D(h))$ such that $M\sqrt{h}\phi(\xi_h) = ch$, that is, $\phi(\xi_h) = c\sqrt{h}/M$. On the difficult branch $\xi_h \rightarrow -\infty$, so this implies $|\xi_h| = O(\sqrt{\log(1/h)})$. Since $|z_D(h) - \sqrt{h} - \xi_h| \leq \sqrt{h}$, the corresponding normal densities are asymptotically equivalent. The lower Mills bound therefore gives, for some constant $C > 0$ and all sufficiently small h , $r_D(h) = M\Phi(z_D(h) - \sqrt{h}) \geq C\sqrt{h}/|\xi_h|$. Hence $r_D(h)/h \geq C/(\sqrt{h}|\xi_h|) \rightarrow +\infty$. Using $M - r_E(h) = ch + r_D(h)$, $q_2(h) = (1 - c)h/(ch + r_D(h)) \rightarrow 0$.

It remains to establish convergence of the supporting multiplier. Fix $1 < \lambda_- < \lambda_T^{\text{NR}} < \lambda_+$. By Lemma A.1, $r_D(h) \rightarrow 0$ and $r_E(h) \rightarrow M$ as $h \downarrow 0$. In the incentive-supply function (B.3), the first and third terms therefore vanish, while the middle term converges to $M\Delta(T, z_1(\lambda, T))$. At $\lambda = \lambda_T^{\text{NR}}$, (B.2) and (A.1) give $z_1(\lambda_T^{\text{NR}}, T) = z_D(T)$. The no-review incentive constraint therefore implies $M\Delta(T, z_1(\lambda_T^{\text{NR}}, T)) = cT$. By (A.1), $\ell_T \in (0, 1)$ and $\lambda_T^{\text{NR}} > 1$. Since $z_1(\cdot, T)$ is strictly increasing and $\Delta(T, \cdot)$ is strictly increasing to the left of $\sqrt{T}/2$, the limiting incentive supply is strictly increasing in λ at λ_T^{NR} . Hence, for all sufficiently small h , $I(\lambda_-; T - h) < c(T - h) < I(\lambda_+; T - h)$. Lemma B.1 then puts every supporting multiplier between λ_- and λ_+ . Since the bracket can be chosen arbitrarily tightly around λ_T^{NR} , $\lambda^*(T - h) \rightarrow \lambda_T^{\text{NR}}$. Q.E.D.

PROOF OF PROPOSITION 2: By Lemma B.1, a supporting multiplier exists. Outside the knife edge it is strictly positive, so the review-score ranking derived in Section 4.2, together with the strict monotonicity of $Q_{\lambda^*}(x)$, gives the ladder structure of a^* : A_B if $x < k_0^*$, A_D if $k_0^* \leq x < k_1^*$, A_E if $k_1^* \leq x < k_2^*$, and A_G if $x \geq k_2^*$. The cutoff formula and the conditions for absent regions follow from (B.1). Since $q_0 < q_1 < q_2$, all finite cutoffs are strictly ordered.

At the knife edge $(1 - c)h = r_D$, the preceding argument applies whenever the optimum is supported by a positive multiplier. If instead $\lambda^* = 0$ supports an optimum, A_B and

A_D are tied at every review score, while A_E and A_G are strictly dominated. Since first-phase work is implemented, an ordered threshold selection between A_B and A_D can be chosen to satisfy obedience. The proposition therefore continues to hold, in this case with $k_1^* = k_2^* = +\infty$. *Q.E.D.*

PROOF OF COROLLARY 1: If $k_0^* = +\infty$, the assigned rent is identically zero and cannot induce first-phase work. Hence $k_0^* < +\infty$. Since Q_{λ^*} is continuous and strictly increasing, the difficult-standard region is also nonempty. Finally, $Q_{\lambda^*}(\mathbb{R}) = (-\infty, \lambda^* - 1)$ reaches $q_1 = 0$ iff $\lambda^* > 1$ and reaches q_2 iff $\lambda^* > 1 + q_2$, which gives the three cases in the corollary. *Q.E.D.*

PROOF OF PROPOSITION 3: Write $h = T - \tau$. By Lemma B.2, $\lambda^*(T - h) \rightarrow \lambda_T^{\text{NR}} > 1$ and $1 + q_2(h) \rightarrow 1$. Hence, for all sufficiently small $h > 0$, $\lambda^*(T - h) > 1 + q_2(h)$. Moreover, $r_D(h)/h \rightarrow \infty$ implies $r_D(h) \neq (1 - c)h$ for all sufficiently small h , so the generic cutoff characterization in Corollary 1 applies. Therefore $k_2^*(T - h) < +\infty$ for every sufficiently small $h > 0$, as claimed. *Q.E.D.*

APPENDIX C: PROOFS FOR SECTION 5

The fixed-review value and the one-sided derivatives

Let $u_D(h) = (1 - c)h - r_D(h)$ and $u_E(h) = (1 - c)h - r_E(h)$ be the principal's payoffs under the two continuation standards, with the argument suppressed where the horizon is fixed. Conditional on first-phase work, the probabilities of the four regions are determined by the standardized cutoffs, and the principal's value from the optimal fixed- τ menu is

$$\begin{aligned} V^R(\tau) = & \tau + u_D \left[\Phi(z_0(\lambda, \tau)) - \Phi(z_1(\lambda, \tau)) \right] \\ & + u_E \left[\Phi(z_1(\lambda, \tau)) - \Phi(z_2(\lambda, \tau)) \right] - M\Phi(z_2(\lambda, \tau)), \end{aligned} \tag{C.1}$$

where $\lambda = \lambda^*(\tau)$ is a supporting multiplier and $\Phi(-\infty) = 0$. Equation (C.1) applies to the two-, three-, and four-region cases under the conventions of (B.2).

For interior review dates, continuity follows from the implicit-function characterization of $z_D(h)$ and $z_E(h)$, continuity of the review-action set, and the maximum theorem applied to the fixed-review linear program. Changes in the number of review regions may create changes in the formula used to compute the value, but not jumps in the value itself.

By Lemma 4 the review gain $G(\tau) = V^R(\tau) - V^{\text{NR}}(T; c, M)$ vanishes at both boundaries, and the natural objects to study there are the one-sided derivatives

$$V^{R'}(0_+) := \lim_{\tau \downarrow 0} \frac{V^R(\tau) - V^{\text{NR}}(T; c, M)}{\tau}, \quad V_-^{R'}(T) := \lim_{\tau \uparrow T} \frac{V^R(\tau) - V^{\text{NR}}(T; c, M)}{\tau - T}, \quad (\text{C.2})$$

whenever the corresponding limits exist. These definitions do not extend V^R to either boundary.

The terminal derivative

Propositions 4(ii) and 6 turn on the sign of $V_-^{R'}(T)$, computed here. Throughout, $z_D(T)$ is the difficult bar of (3) at horizon T , and ℓ_T and λ_T^{NR} are defined in (A.1).

LEMMA C.1: Suppose Assumption 4 holds. Then $V_-^{R'}(T)$ exists, the review value and the no-review value agree to first order at T ,

$$V^R(T - h) = V^{\text{NR}}(T - h; c, M) + o(h), \quad V_-^{R'}(T) = \frac{\partial V^{\text{NR}}(T; c, M)}{\partial T},$$

and

$$V_-^{R'}(T) = 1 - \lambda_T^{\text{NR}} \left[c - \frac{M\phi(z_D(T) - \sqrt{T})}{2\sqrt{T}} \right]. \quad (\text{C.3})$$

Consequently:

(i) $V_-^{R'}(T) < 0$ if and only if

$$\ell_T - (1 - c) > \frac{M\phi(z_D(T) - \sqrt{T})}{2\sqrt{T}}; \quad (\text{C.4})$$

(ii) $V_-^{R'}(T) < 0$ implies $\lambda_T^{\text{NR}} > 1/c$, equivalently $\ell_T > 1 - c$.

PROOF: As $h \downarrow 0$, the work-inducing continuation frontier differs from the no-effort frontier by a vertical surplus of only $(1 - c)h$. Moreover, r_D and $M - r_E$ are both $O(\sqrt{h}/\sqrt{\log(1/h)})$. The continuation actions can therefore improve on the no-effort support function only on likelihood-index intervals whose widths are $O(\sqrt{h \log(1/h)})$. Since the density of the review signal is bounded, their total incremental value is $o(h)$, which gives $V^R(T - h) = V^{\text{NR}}(T - h; c, M) + o(h)$. With $\lim_{\tau \uparrow T} V^R(\tau) = V^{\text{NR}}(T; c, M)$ from Lemma 4 and definition (C.2), this yields $V_-^{R'}(T) = \partial V^{\text{NR}}(T; c, M)/\partial T$.

It remains to differentiate the no-review value. Implicit differentiation of $M[\Phi(z_D(T)) - \Phi(z_D(T) - \sqrt{T})] = cT$ gives

$$\frac{dz_D(T)}{dT} = \frac{c/M - \phi(z_D(T) - \sqrt{T})/(2\sqrt{T})}{\phi(z_D(T)) - \phi(z_D(T) - \sqrt{T})}.$$

Differentiating $V^{\text{NR}}(T; c, M) = T - M\Phi(z_D(T))$ and using (A.1) gives (C.3).

For (i), substitute $\lambda_T^{\text{NR}} = 1/(1 - \ell_T)$ into (C.3) and multiply by $1 - \ell_T > 0$: $V_-^{Rj}(T) < 0$ is equivalent to $c - M\phi(z_D(T) - \sqrt{T})/(2\sqrt{T}) > 1 - \ell_T$, which is (C.4). For (ii), the term $\lambda_T^{\text{NR}} M\phi(z_D(T) - \sqrt{T})/(2\sqrt{T})$ in (C.3) is strictly positive, so $V_-^{Rj}(T) < 0$ forces $1 - \lambda_T^{\text{NR}} c < 0$; and $\lambda_T^{\text{NR}} > 1/c$ is equivalent to $1 - \ell_T < c$. *Q.E.D.*

The left side of (C.4) is large when the reward event is weak evidence of work, that is when the shadow price of terminal incentives is high; the right side is the first-order informational gain from lengthening the horizon. Part (ii) is (C.4) with that right side replaced by zero, and hence weaker. Recall from Lemma 2 that $\underline{M}(T, c) = M_I(T, c)$ when $c \leq \bar{c}(T)$.

LEMMA C.2: Fix $T > 0$ and $c \in (0, 1)$. Then

$$\lim_{M \downarrow M_I(T, c)} V_-^{Rj}(T) = -\infty, \quad \lim_{M \rightarrow \infty} V_-^{Rj}(T) = 1 - c > 0. \quad (\text{C.5})$$

PROOF: Let $a = \sqrt{T}/2$. As $M \downarrow M_I(T, c)$ the two no-review roots merge, so $z_D(T) \uparrow a$, $\ell_T \uparrow 1$, and $\lambda_T^{\text{NR}} \rightarrow \infty$. The bracket in (C.3) converges to $c[1 - a\phi(a)/(2\Phi(a) - 1)]$, which is strictly positive because

$$2\Phi(a) - 1 - a\phi(a) > 0 \quad \text{for every } a > 0. \quad (\text{C.6})$$

Indeed, the expression on the left of (C.6) is zero at $a = 0$ and has derivative $(1 + a^2)\phi(a) > 0$. The first limit in (C.5) follows.

As $M \rightarrow \infty$, $z_D(T) \rightarrow -\infty$ by the proof of Lemma 2; hence $\ell_T \rightarrow 0$, $\lambda_T^{\text{NR}} \rightarrow 1$, and $M\phi(z_D(T) - \sqrt{T}) \rightarrow 0$. Equation (C.3) then gives the second limit. *Q.E.D.*

COROLLARY C.1: Suppose $c \leq \bar{c}(T)$, so that $\underline{M}(T, c) = M_I(T, c)$. Then there exists $\delta > 0$ such that $V_-^{Rj}(T) < 0$ for every $M \in (\underline{M}(T, c), \underline{M}(T, c) + \delta)$.

PROOF: Immediate from the first limit in (C.5). *Q.E.D.*

PROOF OF LEMMA 4: Consider first $\tau \downarrow 0$. The largest principal payoff available from a second-phase action is $u_D = (1-c)h - r_D$ for all sufficiently small τ , because profitability of the no-review contract implies $u_D(T) = V^{\text{NR}}(T; c, M) > 0$. Hence $V^R(\tau) \leq \tau + u_D(T - \tau)$, whose limit is $u_D(T) = V^{\text{NR}}(T; c, M)$.

For a matching lower bound, assign the low-rent continuation action A_D at every review score except on a sufficiently high upper tail, on which the principal assigns the good stop A_G . The utility increment on that tail is $M - r_D$. Choose the upper-tail cutoff so that $[M - r_D] \Delta(\tau, z) = c\tau$. For sufficiently small τ , such a cutoff exists, and the work probability of the reward region converges to zero. The resulting payoff converges to $u_D(T)$. Therefore $\lim_{\tau \downarrow 0} V^R(\tau) = V^{\text{NR}}(T; c, M)$.

Now let $\tau \uparrow T$, so $h \downarrow 0$. A menu using only A_B and A_G is exactly a bounded upper-tail bonus based on X_τ . Choosing its threshold to satisfy $M \Delta(\tau, z) = c\tau$ gives payoff $V^{\text{NR}}(\tau; c, M)$, which converges to $V^{\text{NR}}(T; c, M)$.

For the reverse inequality, the full-effort continuation frontier converges to the no-effort frontier. In particular $(1-c)h \rightarrow 0$ and, by Lemma A.1, $r_D \rightarrow 0$ and $r_E \rightarrow M$. Thus the complete review-action set converges to $\{(r, 0) : r \in [0, M]\}$, so the associated principal payoff converges to $-r$. The review signal converges to the terminal signal $N(T, T)$, and the fixed-review problem converges to the no-review bounded-payment problem. This proves $\lim_{\tau \uparrow T} V^R(\tau) = V^{\text{NR}}(T; c, M)$. *Q.E.D.*

PROOF OF PROPOSITION 4: The two claims concern the one-sided derivatives (C.2). Part (i) establishes that $V^R(0_+) = -\infty$ directly; part (ii) reads the sign of $V_-^R(T)$ off Lemma C.1.

For the first claim, write $h = T - \tau$ and use the low-rent continuation action $A_D(h)$ as a baseline. For all sufficiently small τ , this action uniquely maximizes the principal's post-review payoff. Compactness of the two continuation frontiers and strict profitability at $h = T$ imply that there is a constant $\kappa > 0$ such that every feasible review action (r, s) satisfies $u_D - (s - r) \geq \kappa|r - r_D|$. Let $L_\tau(x) = 1 - g_0^\tau(x)/g_1^\tau(x)$. Because a constant review utility provides no incentives, pre-review obedience can be written as $\mathbb{E}_1[(r(X_\tau) - r_D)L_\tau(X_\tau)] \geq c\tau$. Under work, $X_\tau = \tau + \sqrt{\tau}Z$ and $L_\tau(X_\tau) = 1 - \exp(-\tau/2 - \sqrt{\tau}Z)$. A standard Gaussian tail decomposition gives $\mathbb{E}_1[|r(X_\tau) - r_D|] \geq K\sqrt{\tau}/\sqrt{\log(1/\tau)}$ for some $K > 0$ and all sufficiently small τ . To see the order, split at $|Z| = \sqrt{8\log(1/\tau)}$. On the central event,

$|L_\tau|$ is of order $\sqrt{\tau \log(1/\tau)}$; under both work and shirking, the complementary Gaussian tails contribute $o(\tau)$. Since review utilities are bounded by M , $\mathbb{E}_1 [(r(X_\tau) - r_D) L_\tau(X_\tau)] \geq c\tau$ then implies $\mathbb{E}_1 [r(X_\tau) - r_D] \geq K\sqrt{\tau}/\sqrt{\log(1/\tau)}$.

Combining $u_D - (s - r) \geq \kappa|r - r_D|$ and $\mathbb{E}_1 [r(X_\tau) - r_D] \geq K\sqrt{\tau}/\sqrt{\log(1/\tau)}$ yields $V^R(\tau) \leq \tau + u_D(T - \tau) - \kappa K\sqrt{\tau}/\sqrt{\log(1/\tau)}$. Since $\tau + u_D(T - \tau) = V^{\text{NR}}(T; c, M) + O(\tau)$, it follows that

$$\lim_{\tau \downarrow 0} \frac{V^R(\tau) - V^{\text{NR}}(T; c, M)}{\tau} = -\infty. \quad (\text{C.7})$$

Thus $V^R(\tau) < V^{\text{NR}}(T; c, M)$ for all sufficiently small $\tau > 0$. Since the boundary limit established above gives $V^R(s) \rightarrow V^{\text{NR}}(T; c, M)$ as $s \downarrow 0$, each such τ is strictly dominated by some genuine review date $s \in (0, \tau)$. This proves part (i).

For the second claim, let M lie in the right neighborhood supplied by Corollary C.1, so that $V_-^{R'}(T) < 0$. By Lemma C.1, $V^R(T - h) = V^{\text{NR}}(T; c, M) - V_-^{R'}(T)h + o(h)$. Comparing the dates $\tau = T - h$ and $\tau' = T - 2h$, $V^R(\tau') - V^R(\tau) = -V_-^{R'}(T)h + o(h) > 0$ for all sufficiently small $h > 0$. Hence every sufficiently late review is strictly dominated by an earlier genuine review, which proves the second claim. *Q.E.D.*

PROOF OF PROPOSITION 5: Write $R_M(t) \equiv R^{\text{NR}}(t; c, M) = r_D(t)$. By Lemma 2, $R_M(T) \rightarrow 0$ as $M \rightarrow \infty$, so the no-review full-effort contract is feasible and profitable for all sufficiently large M . Suppose toward a contradiction that there are sequences $M_n \rightarrow \infty$ and $\tau_n \in (0, T)$ such that

$$V^R(\tau_n) \geq V^{\text{NR}}(T; c, M_n). \quad (\text{C.8})$$

Dependence on n is suppressed below. Any continuation-utility schedule $r(X_\tau) \in [0, M]$ that induces first-phase work is itself a bounded incentive contract for a phase of length τ . By the definition of $R_M(\tau)$, the agent's ex ante rent under the review contract is therefore at least $R_M(\tau)$. Total surplus is at most $(1 - c)T$, and hence $V^R(\tau) \leq (1 - c)T - R_M(\tau)$. Comparison with (C.8) yields the necessary condition $R_M(\tau) \leq R_M(T)$.

We first show that any such sequence must satisfy $\tau_n \log M_n \rightarrow 0$. Implicit differentiation of the binding difficult-root condition gives

$$R'_M(t) = \frac{\phi(z_D(t) - \sqrt{t})}{\phi(z_D(t)) - \phi(z_D(t) - \sqrt{t})} \left[c - \frac{M\phi(z_D(t))}{2\sqrt{t}} \right]. \quad (\text{C.9})$$

For every $\varepsilon > 0$, $z_D(t) \rightarrow -\infty$ uniformly on $t \in [\varepsilon, T]$ as $M \rightarrow \infty$. The Mills bound $\Phi(z) \leq \phi(z)/|z|$ for $z < 0$, together with $M\Delta(t, z_D(t)) = ct$, makes the bracket in (C.9) negative uniformly on that interval. The prefactor is positive on the difficult branch. Hence $R'_M(t) < 0$ on $[\varepsilon, T]$ for all sufficiently large M . Condition $R_M(\tau) \leq R_M(T)$ then forces $\tau_n \rightarrow 0$.

We sharpen this localization. Let $a_M < 0$ solve $\Phi(a_M) = 2cT/M$, and put $q_M = \Phi(a_M - \sqrt{T})/\Phi(a_M)$. The threshold bonus $M/[2(1 - q_M)]$ is feasible for large M , implements effort, and leaves rent $cTq_M/(1 - q_M)$. Gaussian tail bounds therefore give

$$R_M(T) \leq 2cT \exp\left(-\sqrt{T \log M} - \frac{T}{2}\right) \quad (\text{C.10})$$

for all sufficiently large M .

For a short phase of length τ , write $z_D(\tau) = -x$ and $u = \sqrt{\tau}$. The binding condition implies $R_M(\tau) = c\tau q/(1 - q) \geq c\tau q$, where $q = \Phi(-(x + u))/\Phi(-x)$. The two-sided Mills inequalities and $x \leq \sqrt{2 \log(M/(c\tau))}$ give

$$R_M(\tau) \geq \frac{c\tau}{2(1 + \sqrt{T})} \exp\left[-\sqrt{2\tau \log\left(\frac{M}{c\tau}\right)} - \frac{\tau}{2}\right]. \quad (\text{C.11})$$

If $\tau_n \log M_n$ were bounded away from zero along a subsequence, then the logarithm of the right side of (C.11) would be $-o(\sqrt{\log M_n})$, whereas (C.10) is of order $-\sqrt{T \log M_n}$. It would follow that $R_{M_n}(\tau_n) > R_{M_n}(T)$, contradicting $R_M(\tau) \leq R_M(T)$. This proves $\tau_n \log M_n \rightarrow 0$.

It remains to show that a review in this vanishingly early region is also strictly dominated. Put $h = T - \tau$, $r_0 = R_M(h)$, and let the difficult continuation action be (r_0, s_0) , where $s_0 = (1 - c)h$ is total surplus and its principal payoff is $s_0 - r_0$. Uniformly for $h \in [T/2, T]$, $r_0 \rightarrow 0$; hence $2r_0 < (1 - c)h$ for all sufficiently large M . Every continuation action $(r, s) \in \mathcal{A}(h)$ then satisfies $s - r \leq s_0 - r_0 - |r - r_0|$. For a work-inducing action $s = (1 - c)h = s_0$ and $r \geq r_0$, so the inequality holds with equality. For a no-effort action $s = 0$; it follows immediately when $r \geq r_0$, and when $r < r_0$ it follows from $2r_0 < (1 - c)h$.

Let $d(x) = r(x) - r_0$. Integrating $s - r \leq s_0 - r_0 - |r - r_0|$ under first-phase work gives

$$V^R(\tau) \leq \tau + s_0 - r_0 - \mathbb{E}_1[d(X_\tau)]. \quad (\text{C.12})$$

Since a constant rent contributes nothing to incentives, first-phase obedience is

$$\mathbb{E}_1[d(X_\tau)L_\tau(X_\tau)] \geq c\tau, \quad (\text{C.13})$$

Under work, $X_\tau = \tau + \sqrt{\tau}Z$ and $L_\tau(X_\tau) = 1 - \exp(-\tau/2 - \sqrt{\tau}Z)$. On the event $|L_\tau| \leq 1/2$, the left side of (C.13) is bounded by $\mathbb{E}_1|d|/2$. On its complement, $|d| \leq M$, and Gaussian tail bounds give, with $a = \log(3/2)$, $\mathbb{E}_1[|L_\tau|\mathbf{1}_{\{|L_\tau| > 1/2\}}] \leq 4\exp(-a^2/(8\tau))$. Because $\tau_n \log M_n \rightarrow 0$ is equivalent to $\log M = o(1/\tau)$, this tail contribution is $o(\tau)$. Thus

$$\mathbb{E}_1|d(X_\tau)| \geq 2c\tau - o(\tau) > c\tau \quad (\text{C.14})$$

for all sufficiently large n .

Finally, subtracting $V^{\text{NR}}(T; c, M) = (1 - c)T - R_M(T)$ from (C.12) yields

$$V^R(\tau) - V^{\text{NR}}(T; c, M) \leq c\tau + R_M(T) - R_M(T - \tau) - \mathbb{E}_1|d(X_\tau)|.$$

For large n , $T - \tau_n \in [T/2, T)$ and the monotonicity established after (C.9) gives $R_M(T - \tau) > R_M(T)$. Combining this strict inequality with (C.14) makes the displayed difference negative, contradicting (C.8). No such sequence exists, so a finite uniform $\bar{M}(T, c)$ exists.

Q.E.D.

PROOF OF PROPOSITION 6: By Lemma 4, $G(\tau) \rightarrow 0$ as $\tau \uparrow T$, and $V_-^{Rl}(T)$ in (C.2) is exactly the left derivative of G at T : $G(\tau) = V_-^{Rl}(T)(\tau - T) + o(T - \tau)$ as $\tau \uparrow T$. Under the stated hypotheses, Corollary C.1 gives $V_-^{Rl}(T) < 0$. Since $\tau - T < 0$, the leading term is strictly positive, so there is $\delta > 0$ with $G(\tau) > 0$ for every $\tau \in (T - \delta, T)$. *Q.E.D.*

The same display gives the converse: if $V_-^{Rl}(T) > 0$ then $G(\tau) < 0$ throughout a left neighborhood of T .

PROOF OF PROPOSITION 7: At the boundary, $\bar{\Delta}(T)/T = c/M_0$. Since $\bar{\Delta}(t)/t$ is strictly decreasing by Lemma 2, every $t < T$ satisfies $\bar{\Delta}(t)/t > c/M_0$, which gives $M_0\bar{\Delta}(\tau) > c\tau$ and $M_0\bar{\Delta}(h) > ch$.

To obtain the second claim, choose $\tau_0 > 0$ small and write $h_0 = T - \tau_0$. At $M = M_0$ the continuation problem at $h_0 < T$ is strictly feasible, so its minimum work-inducing rent $r_D(h_0)$ is well defined and strictly below M_0 . At the boundary $h = T$, the two threshold

roots coincide and the corresponding rent is $M_0\Phi(-\sqrt{T}/2)$; hence the available rent spread converges to $M_0\Phi(\sqrt{T}/2) > 0$ as $h_0 \uparrow T$. Since $\bar{\Delta}(\tau)/\tau \rightarrow \infty$ as $\tau \downarrow 0$, τ_0 can therefore be chosen small enough that $[M_0 - r_D(h_0)]\bar{\Delta}(\tau_0) > c\tau_0$. Both continuation feasibility at h_0 and the above inequality are strict, so by continuity they continue to hold for all caps in a sufficiently small interval $(M_0 - \varepsilon, M_0]$.

For such a cap, use the minimum-rent full-effort continuation contract after review scores $X_{\tau_0} < \tau_0/2$, and use the good-stop action with rent M after $X_{\tau_0} \geq \tau_0/2$. The spread in continuation utility between the two regions is $M - r_D(h_0)$, and the midpoint cutoff maximizes the difference between the work and shirk probabilities. The date-0 incentive spread is therefore $[M - r_D(h_0)]\bar{\Delta}(\tau_0) > c\tau_0$, so first-phase work is induced. In the lower region the continuation contract induces full effort, while in the upper region the project is stopped. The lower region has strictly positive probability under work. Finally, $M < M_0 = M_I(T, c)$ makes full effort infeasible without review by the implementability condition in Lemma 2. *Q.E.D.*

APPENDIX D: PROOFS FOR SECTION 6

PROOF OF PROPOSITION 8: Let $W_t = \mathbb{E}_t[Y - c \int_t^T e_s ds]$ be promised utility. Martingale representation gives $dW_t = ce_t dt + \beta_t(dX_t - e_t dt)$ with $W_T = Y$. A local deviation from work to shirk saves c and reduces the drift of promised utility by β_t , so work requires $\beta_t \geq c$. Also $0 \leq W_t \leq M$: permanent shirking guarantees a nonnegative continuation payoff, whereas $Y \leq M$. At $W_t = 0$, incentive compatibility forces both the terminal payment and all subsequent effort to be zero. At $W_t = M$, the identity $M = \mathbb{E}_t Y - c\mathbb{E}_t \int_t^T e_s ds \leq M - c\mathbb{E}_t \int_t^T e_s ds$ gives the same conclusion, with terminal payment M . Thus both boundaries are absorbing.

Scale promises by c : $u = w/c$, $m = M/c$, and let $v(h, u)$ be maximal expected remaining work. Because $\mathbb{E}Y = W_0 + c\mathbb{E} \int_0^T e_t dt$, the principal's payoff is $(1 - c)\mathbb{E} \int_0^T e_t dt - W_0$. With $b = \beta/c$, the recursive equation is

$$v_h = \max \left\{ \sup_{b \geq 1} [1 + v_u + \frac{1}{2}b^2 v_{uu}], \sup_{b \leq 1} \frac{1}{2}b^2 v_{uu} \right\},$$

with $v(h, 0) = v(h, m) = v(0, u) = 0$. Under the candidate policy the agent works in the interior with $b = 1$, so U satisfies $dU = dt + dB$ until its first exit σ from $(0, m)$ and

$v(h, u) = \mathbb{E}_u[\sigma \wedge h]$. Hence $v_h = 1 + v_u + \frac{1}{2}v_{uu}$. The parabolic maximum principle applied to v_h and v_{uu} gives $v_h \geq 0$ and $v_{uu} \leq 0$. The work branch is therefore maximized at the smallest admissible b , namely 1, while the shirk branch is at most zero and cannot exceed v_h . Verification yields the band policy. Returning to dollars gives $dW_t = c dX_t$, hence $W_t = W_0 + cX_t$ until a boundary is hit. Finally the principal chooses $u = W_0/c$ to maximize $(1 - c)v(T, u) - cu$; concavity of v gives uniqueness whenever the optimum is interior. *Q.E.D.*

PROOF OF PROPOSITION 9: Before τ_1 the agent learns nothing about the realized review date, so a deviation is summarized by cumulative effort $e \in [0, \tau_1]$. If the early review occurs his payoff is $U_1(e)$; otherwise he learns that the date is τ_2 and reoptimizes over cumulative effort in $[e, e + d]$, obtaining $\hat{U}_2(e)$. Sequential optimality after non-arrival therefore requires $U_2(\tau_2) = \hat{U}_2(\tau_1)$. Given this, full effort before τ_1 is optimal iff $pU_1(\tau_1) + (1 - p)U_2(\tau_2) \geq pU_1(e) + (1 - p)\hat{U}_2(e)$ for every e , which is (11). This proves part (i).

If the late branch induces full effort on its own, then $D_2(e) \geq 0$. Where $D_1(e) \geq 0$ the constraint holds for every p ; where $D_1(e) < 0$ it is equivalent to $p \leq D_2(e)/[D_2(e) - D_1(e)]$. Taking the infimum gives (12), proving part (ii). Under full effort the lottery payoff is $P_2 + p(P_1 - P_2)$. When $P_1 > V^{D, \text{pw}} > P_2$, this exceeds $V^{D, \text{pw}}$ iff $p > \underline{p}$, whereas sequential incentive compatibility requires $p \leq \bar{p}$. Thus a feasible strict improvement exists iff $\underline{p} < \bar{p}$, and every p in the stated interval works. *Q.E.D.*

REFERENCES

- Baker, George P., Michael C. Jensen, and Kevin J. Murphy (1988), “Compensation and incentives: Practice vs. theory.” *The Journal of Finance*, 43 (3), 593–616. [2]
- Ball, Ian and Jan Knoepfle (2026), “Should the timing of inspections be predictable?”, URL <https://arxiv.org/abs/2304.01385>. Working paper, available at arXiv:2304.01385. [7]
- Barlevy, Gadi and Derek Neal (2019), “Allocating effort and talent in professional labor markets.” *Journal of Labor Economics*, 37 (1), 187–246. [2]
- Cappelli, Peter and Martin J. Conyon (2018), “What do performance appraisals do?” *ILR Review*, 71 (1), 88–116. [2]
- Chen, Bin R. and Y. Stephen Chiu (2013), “Interim performance evaluation in contract design.” *The Economic Journal*, 123 (569), 665–698. [5]
- Dai, Liang, Yenan Wang, and Ming Yang (2026), “Dynamic contracting with flexible monitoring.”, URL <https://ssrn.com/abstract=3496785>. Available at SSRN 3496785. [7]

- 1 Deserranno, Erika, Philipp Kastrau, and Gianmarco León-Ciliotta (2025), “Promotions and productivity: The role 1
2 of meritocracy and pay progression in the public sector.” *American Economic Review: Insights*, 7 (1), 71–89. [2] 2
- 3 DeVaro, Jed and Dana Brookshire (2007), “Promotions and incentives in nonprofit and for-profit organizations.” 3
4 *ILR Review*, 60 (3), 311–339. [2] 4
- 5 Ely, Jeffrey C., George Georgiadis, and Luis Rayo (2025), “Feedback design in dynamic moral hazard.” *Economet-* 5
6 *rica*, 93 (2), 597–621. [6] 5
- 7 Ely, Jeffrey C. and Martin Szydlowski (2020), “Moving the goalposts.” *Journal of Political Economy*, 128 (2), 6
8 468–506. [5] 7
- 9 Georgiadis, George and Balázs Szentes (2020), “Optimal monitoring design.” *Econometrica*, 88 (5), 2075–2107. 8
10 [7] 9
- 11 Hoffmann, Florian, Roman Inderst, and Marcus M. Opp (2021), “Only time will tell: A theory of deferred 10
12 compensation.” *The Review of Economic Studies*, 88 (3), 1253–1278. [7] 10
- 13 Holmström, Bengt (1979), “Moral hazard and observability.” *The Bell Journal of Economics*, 10 (1), 74–91. [15] 11
- 14 Jewitt, Ian, Ohad Kadan, and Jeroen M. Swinkels (2008), “Moral hazard with bounded payments.” *Journal of* 12
15 *Economic Theory*, 143 (1), 59–82. [5] 13
- 16 Levin, Jonathan and Steven Tadelis (2005), “Profit sharing and the role of professional partnerships.” *The Quarterly* 14
17 *Journal of Economics*, 120 (1), 131–171. [2] 14
- 18 Liu, Yijun (2026), “Resource allocation with midterm review.” Working paper. [5] 15
- 19 Lizzeri, Alessandro, Margaret A. Meyer, and Nicola Persico (2002), “The incentive effects of interim performance 16
20 evaluations.” CARESS Working Paper 02-09, University of Pennsylvania. [5] 17
- 21 Lukas, Christian (2010), “Optimality of intertemporal aggregation in dynamic agency.” *Journal of Management* 18
22 *Accounting Research*, 22 (1), 157–174. [5] 18
- 23 Mas, Alexandre (2017), “Does transparency lead to pay compression?” *Journal of Political Economy*, 125 (5), 19
24 1683–1721. [2] 20
- 25 Ray, Korok (2007), “Performance evaluations and efficient sorting.” *Journal of Accounting Research*, 45 (4), 21
26 839–882. [5] 22
- 27 Rodivilov, Alexander (2022), “Monitoring innovation.” *Games and Economic Behavior*, 135, 297–326. [7] 23
- 28 Sannikov, Yuliy (2008), “A continuous-time version of the principal–agent problem.” *The Review of Economic* 24
29 *Studies*, 75 (3), 957–984. [6, 26] 24
- 30 Spear, Stephen E. and Cheng Wang (2005), “When to fire a CEO: Optimal termination in dynamic contracts.” 25
31 *Journal of Economic Theory*, 120 (2), 239–256. [6] 26
- 32 Varas, Felipe, Iván Marinovic, and Andrzej Skrzypacz (2020), “Random inspections and periodic reviews: Optimal 27
dynamic monitoring.” *The Review of Economic Studies*, 87 (6), 2893–2937. [6] 28
- Wong, Yu Fu (2026), “Dynamic monitoring design.”, URL <https://ssrn.com/abstract=4466562>. Working paper, 29
available at SSRN 4466562. [7] 29
- Zhu, John Y. (2013), “Optimal contracts with shirking.” *The Review of Economic Studies*, 80 (2), 812–839. [6] 30
31 31
32 32